

Basic Principle of Medicine 1
**Module: Cardiovascular, Pulmonary
and Renal**
Lecture: CPR 12

Cardiac Action Potential

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LEARNING OBJECTIVES

- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0164** Compare and contrast the ventricular myocyte action potential and the pacemaker action potential in terms of their phases and ionic basis.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0165** Explain the role of ion channels that contribute to various phases of the action potential of the ventricular myocytes and the pacemaker cells.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0166** Explain the causes of the long duration of the cardiac action potential, the resultant long refractory period, and the advantages of these characteristics for the cardiac muscle
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0167** Beginning in the SA node, diagram the normal sequence of cardiac activation (depolarization).
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0168** Explain why the AV node is the only electrical pathway between the atria and the ventricles, the functional importance of the slow conduction through the AV node, and the factors that influence conduction velocity through the AV node
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0169** Explain the characteristics of the pacemaker cells of the heart, the ionic mechanism of automaticity and rhythmicity, their location and spontaneous rate, and the neural and humoral factors that influence their rate
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0170** Compare the sympathetic and parasympathetic nervous system influence on heart rate and cardiac excitation and which division is dominant at rest and during exercise.



Outline

Electrical potential, resting membrane potential, depolarization and repolarization, threshold potential

Cardiac Action Potential

- Cardiac muscle cell action potential
- Pacemaker Cell Action Potential

Ions channels and cardiac action potential

Refractory period

Sequence of cardiac depolarization

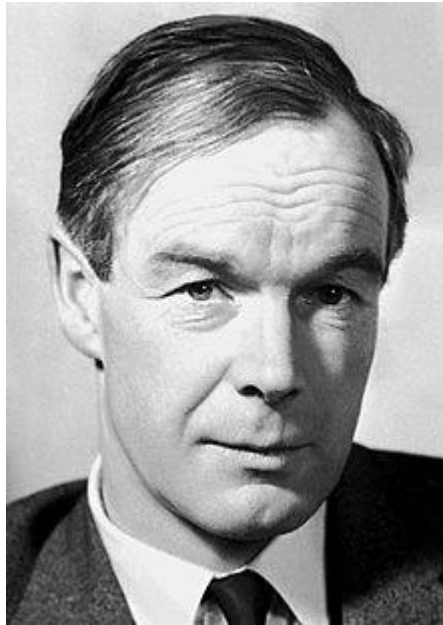
A-V Node characteristics

Autonomic Nervous System influence on heart rate and cardiac excitation

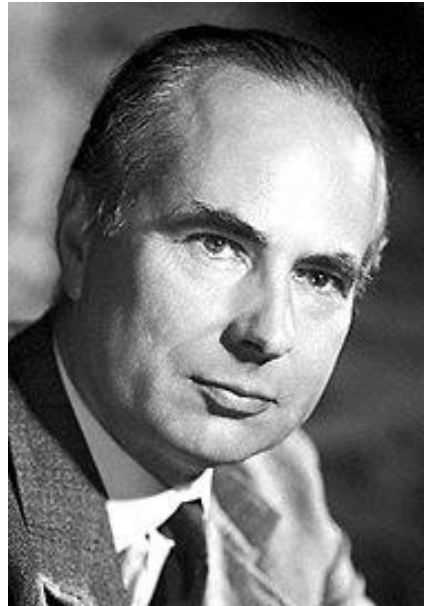


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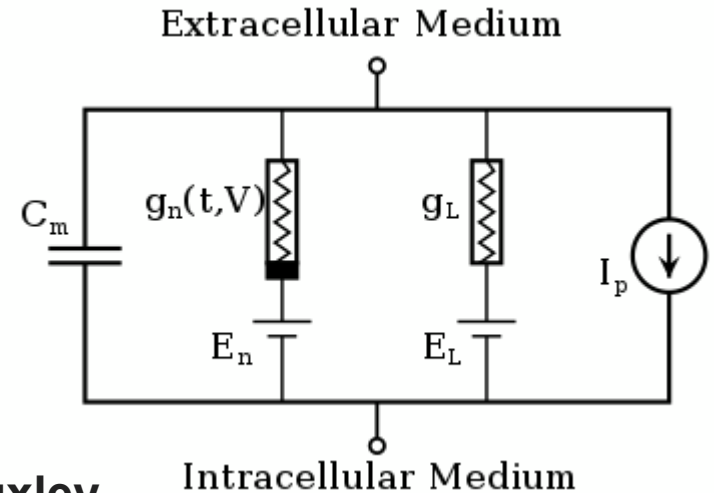


Sir Alan Lloyd Hodgkin



Sir Andrew Fielding Huxley

Hodgkin-Huxley model (conductance base model) 1952



**Nobel Prize in Physiology / Medicine
1963**

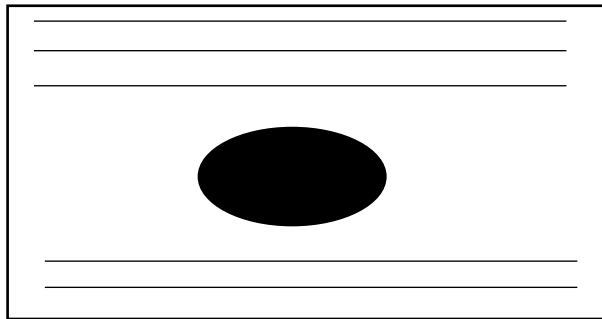
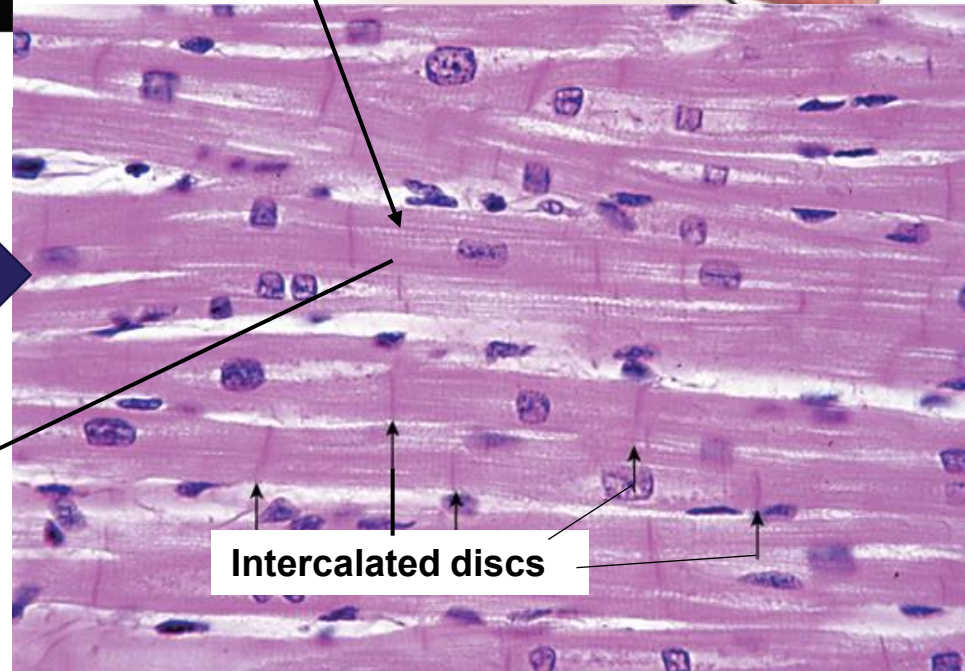
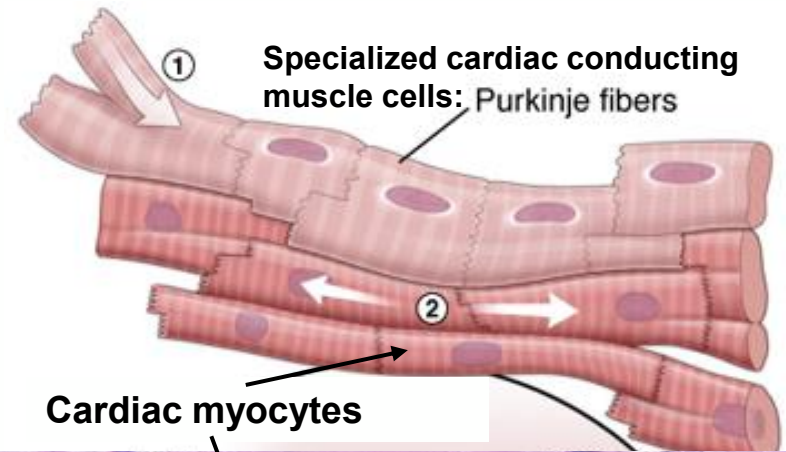


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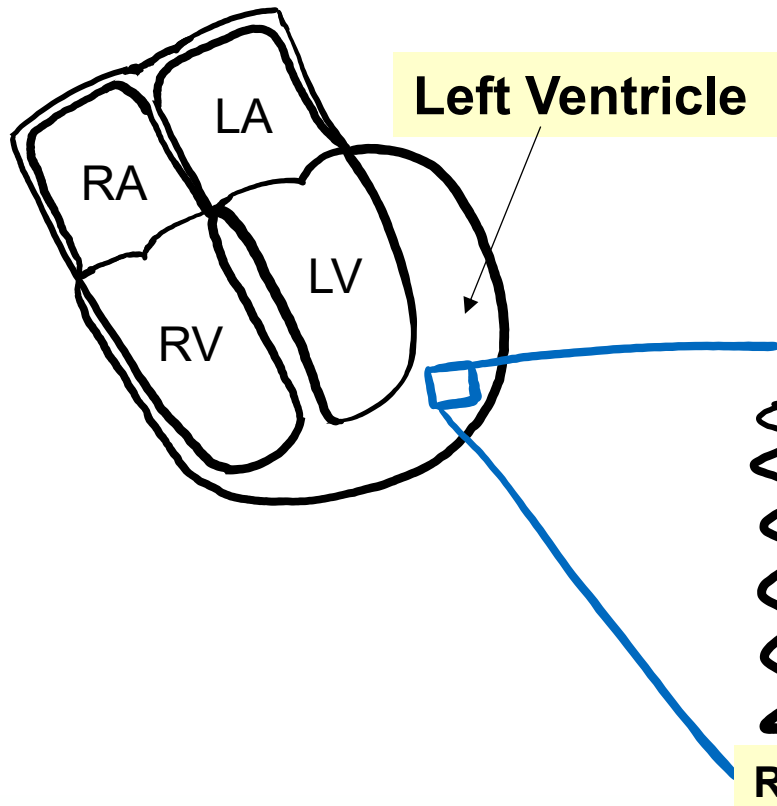


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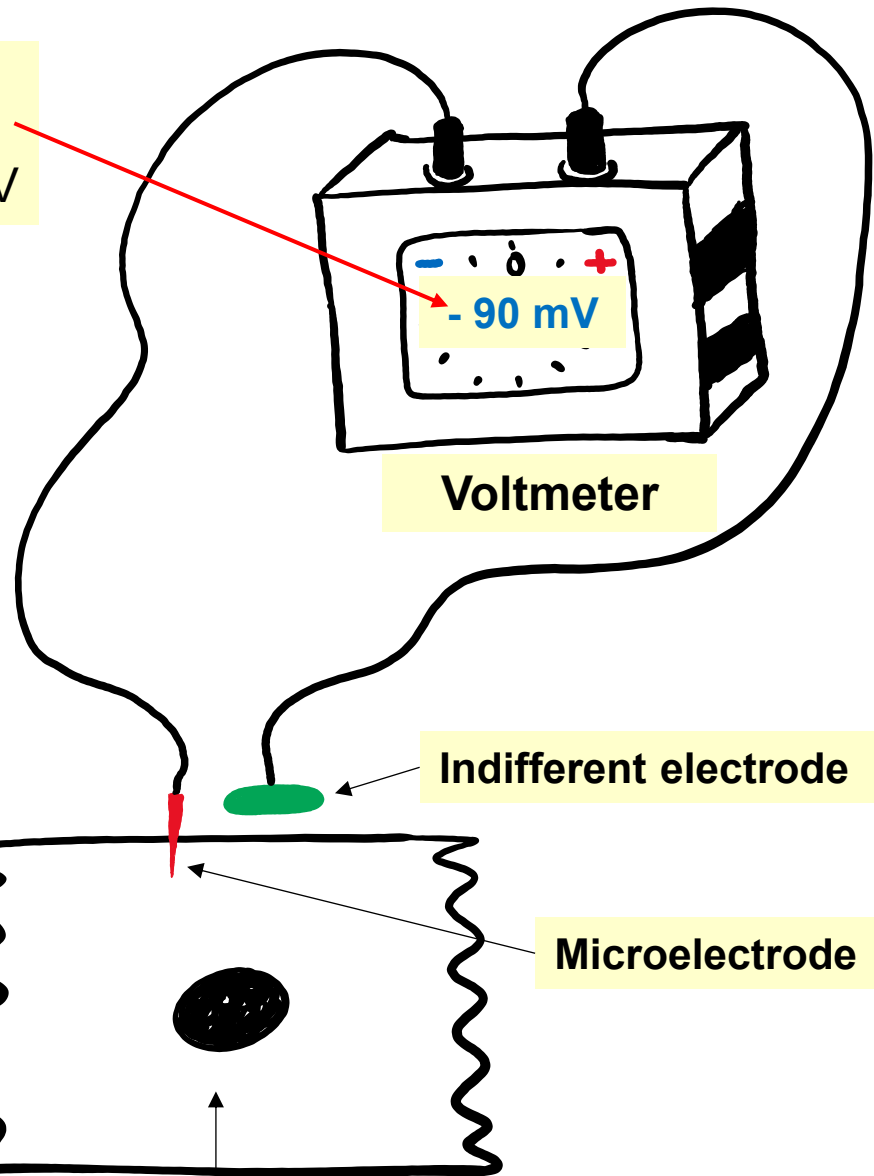
Resting Membrane Potential
of the cardiomyocyte is -90 mV

Heart



Left Ventricle

Resting ventricular myocyte



Voltmeter

Indifferent electrode

Microelectrode



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What means that the **CARDIOMYOCYTE** has a **Resting Membrane Potential of – 90 mV ?**

The potential inside the cardiomyocyte is 90 mV more negative than the potential in the extracellular fluid on the outside of the cardiomyocyte

What determines the Membrane Potential?

Concentration of (+) and (-) charged IONS across the membrane

The relative permeability of the cell membrane to these IONS

Ionic pumps that transport IONS across the membrane

K^+ , Na^+ , Ca^{++} , Cl^-



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ION CONCENTRATION INSIDE AND OUTSIDE OF RESTING MYOCYTE

ION	INSIDE	OUTSIDE mEq/L
K ⁺	140 mEq/L	4 mEq/L
Na ⁺	10 mEq/L	142 mEq/L
Ca ⁺⁺	0.0001 mEq/L	2.4 mEq/L
Cl ⁻	4 mEq/L	103 mEq/L

Concentration ratios:

$$K^+ (\text{inside}) / K^+ (\text{outside}) = 35.0$$

$$Na^+ (\text{inside}) / Na^+ (\text{outside}) = 0.1$$

Why we have those large concentration gradients of K⁺ and Na⁺ across the cell membrane?

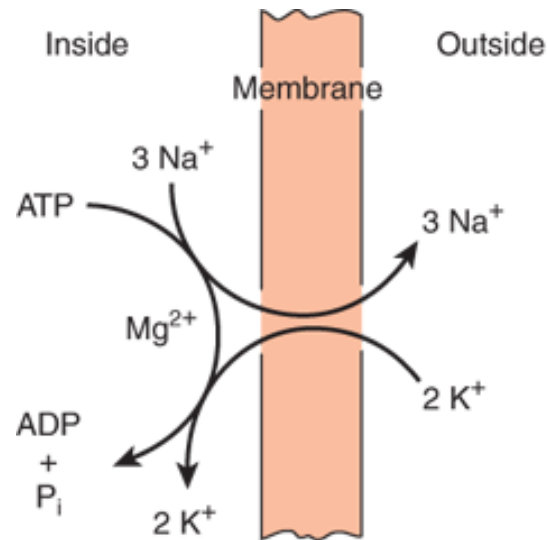


The Sodium-Potassium Pump (ATPase)

3 Na⁺ to the outside for each 2 K⁺ to the inside

CONCENTRATION GRADIENT INSIDE/OUTSIDE

ELECTROGENIC PUMP



Source: V.W. Rodwell, D.A. Bender, K.M. Botham, P.J. Kennelly
P.A. Weil: Harper's Illustrated Biochemistry, 31st Edition
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Action Potentials in the Heart



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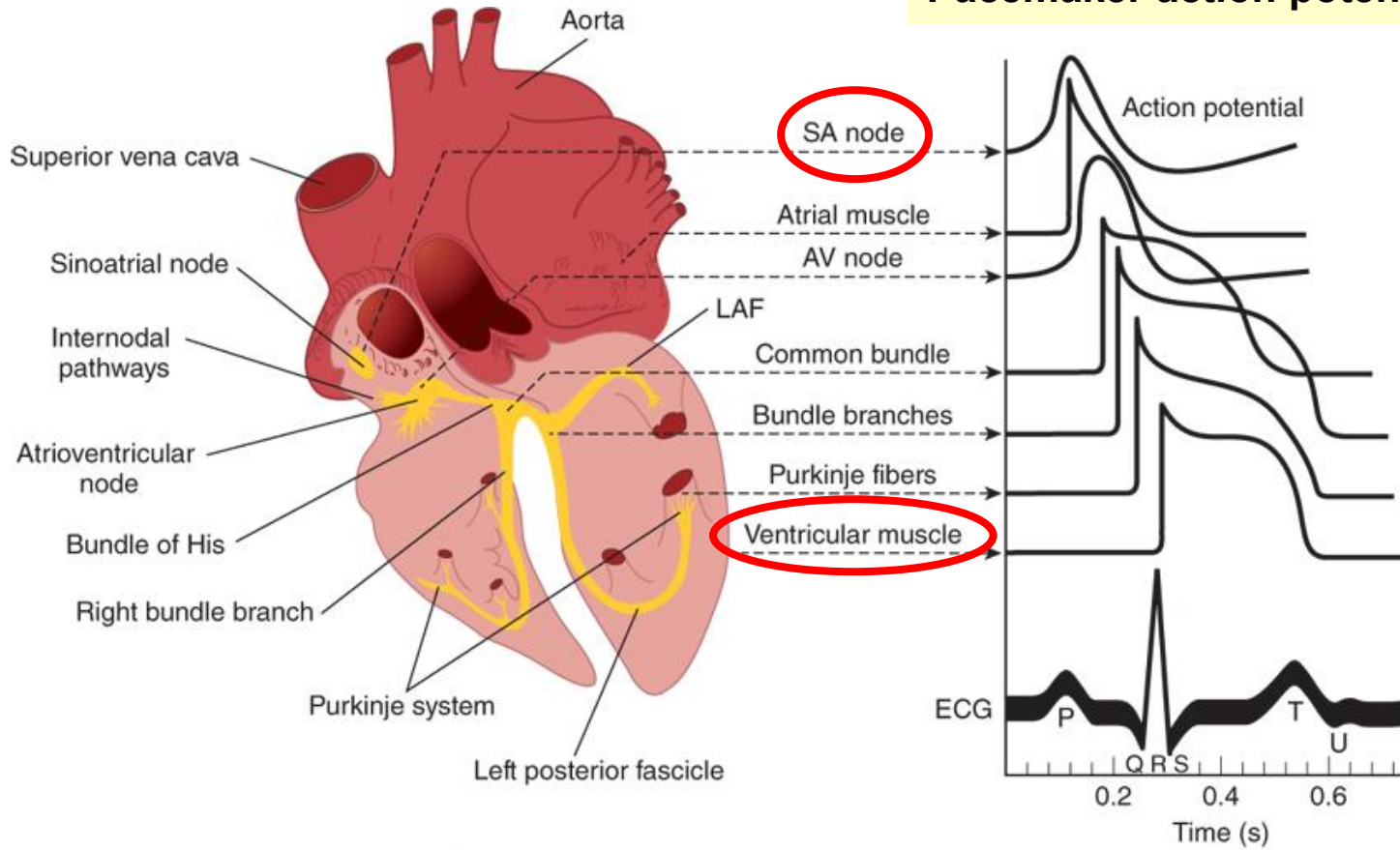
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Cardiac Action Potential

Non-pacemaker action potential:

- Atrial myocyte
- Ventricular myocyte
- Purkinje fiber

Pacemaker action potential



Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan:
Ganong's Review of Medical Physiology, Twenty-Sixth Edition
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ION CHANNELS AND CARDIAC ACTION POTENTIAL



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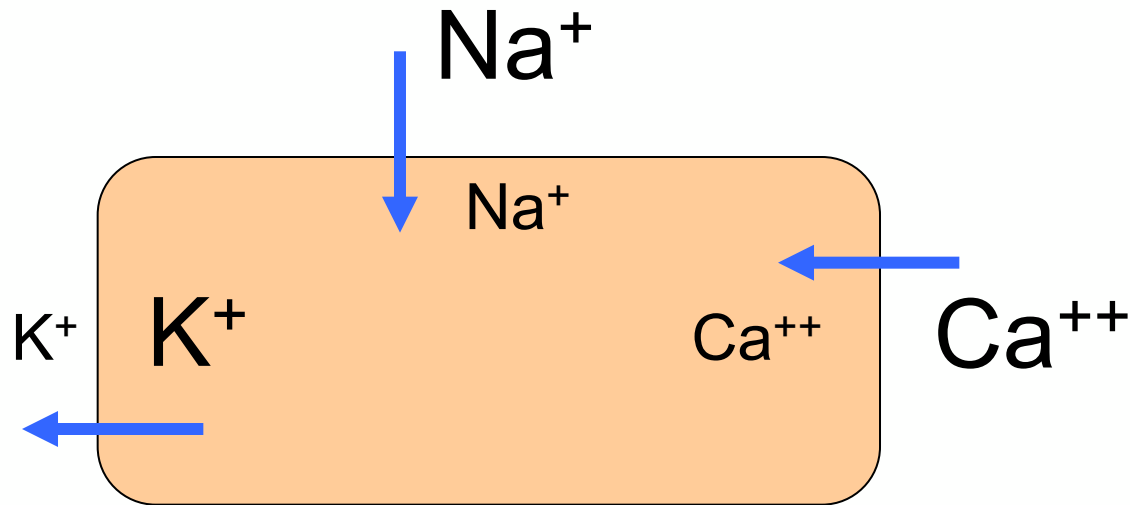
Cardiac Ion Channels and Currents

Channel	Gating	Characteristics
<u>Sodium</u>		
Fast Na ⁺ (I_{Na})	Voltage	Phase 0 of myocytes
Slow Na ⁺ (I_f)	Voltage and receptor	Contributes to phase 4 pacemaker current in SA and AV nodal cells
<u>Calcium</u>		
L-type (I_{Ca})	Voltage	Slow inward, long lasting current; phase 2 of myocytes and phases 4 and 0 of SA and AV nodal cells
T-type (I_{Ca})	Voltage	Transient current; contributes to phase 4 pacemaker current in SA and AV nodal cell
<u>Potassium</u>		
Inward rectifier (I_{K1})	Voltage	Maintains negative potential in phase 4: closes with depolarization; its decay contributes pacemaker currents
Transient outward (I_{to})	Voltage	Contributes to phase 1 in myocytes
Delayed rectifier (I_{Kr})	Voltage	Phase 3 repolarization
ATP-sensitive (I_K , ATP)	Receptor	Inhibited by ATP; opens when ATP decreases
Acetylcholine activated (I_K , ACh)	Receptor	Activated by acetylcholine; Gi-protein coupled



Cardiac Action Potential

The Ionic Gradients across the Ventricular Myocyte



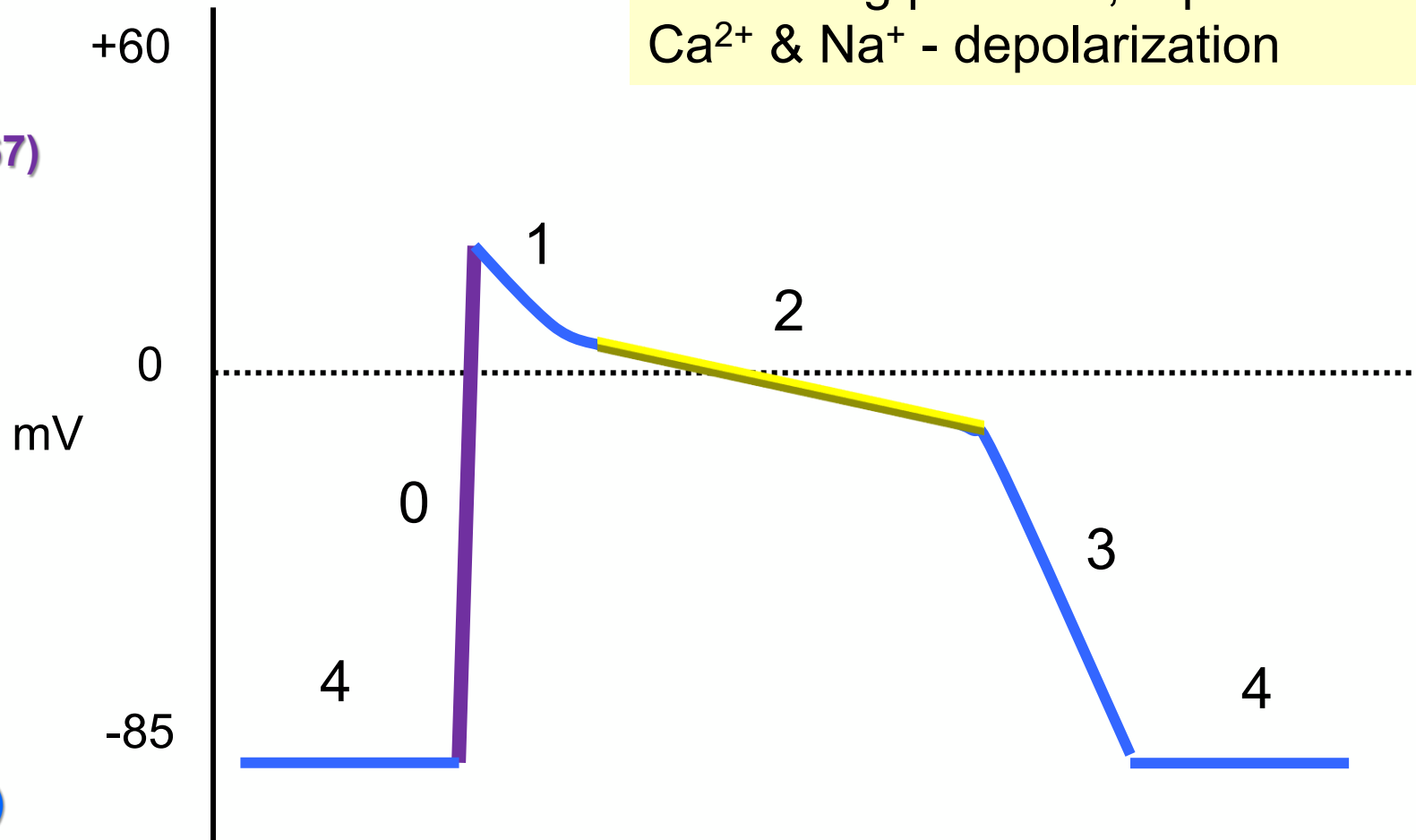
Direction of current flow is determined by the electrochemical gradient for that ion AND conductance. * Note: there are no large changes in ionic concentrations.

Typical Cardiac Action Potential from a Ventricular Myocyte

Ca^{2+} (+123)

K^+ - resting potential, repolarization
 Ca^{2+} & Na^+ - depolarization

Na^+ (+67)



K^+ (-92)

Eq potential
of ions

250 – 300 ms



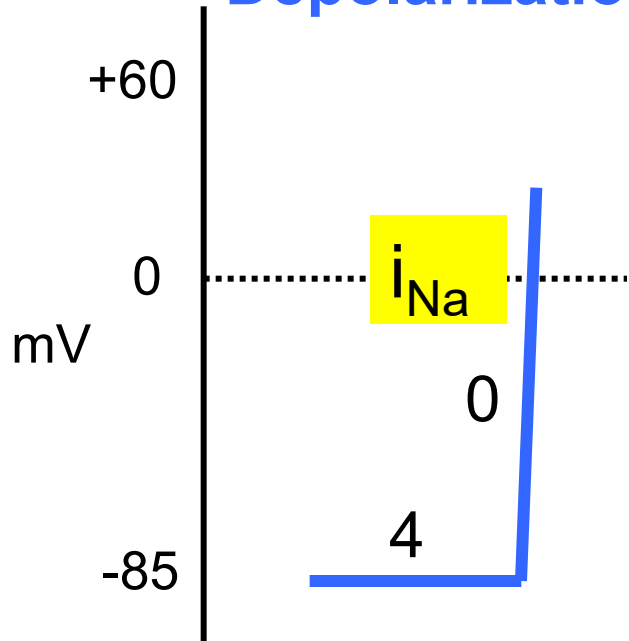
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Cardiac Action Potential from a Ventricular Myocyte Cell

Phase 0

**Rapid
Depolarization**



Depolarization $>$ threshold triggers opening of voltage-gated Na^+ channels (rapid opening of activation or m gate opens).

Na^+ rapidly enters cell

Cell depolarizes, inside becomes +ve and approaches E_{Na}

Two events prevent depolarization to E_{Na}

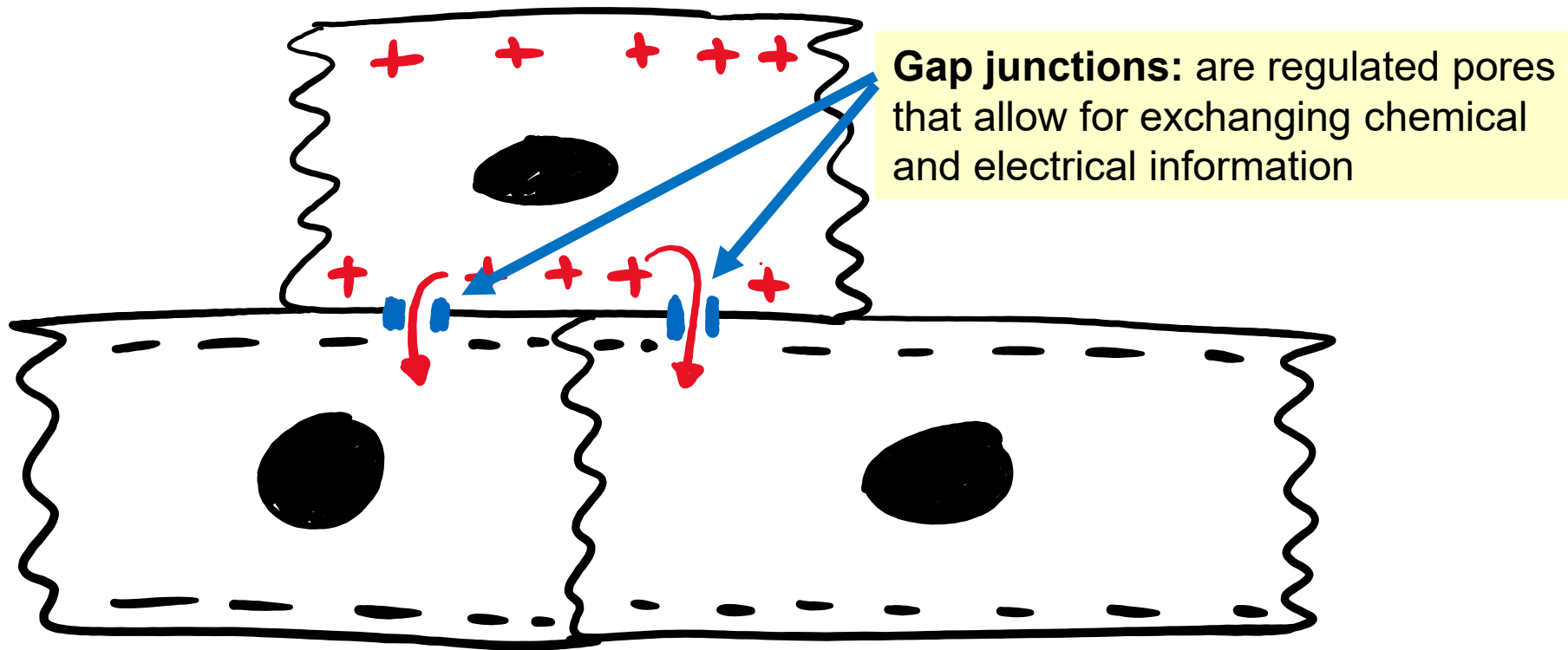
Na inactivation process starts (inact. or h gate closes as V_m less neg)

Membrane starts to repolarize

Depolarization of the cell triggers the opening of voltage-gated K^+ channels (repolarization)

Outward flow of K^+ i_{to}

ACTION POTENTIAL CONDUCTED BY AN ADJACENT CELL



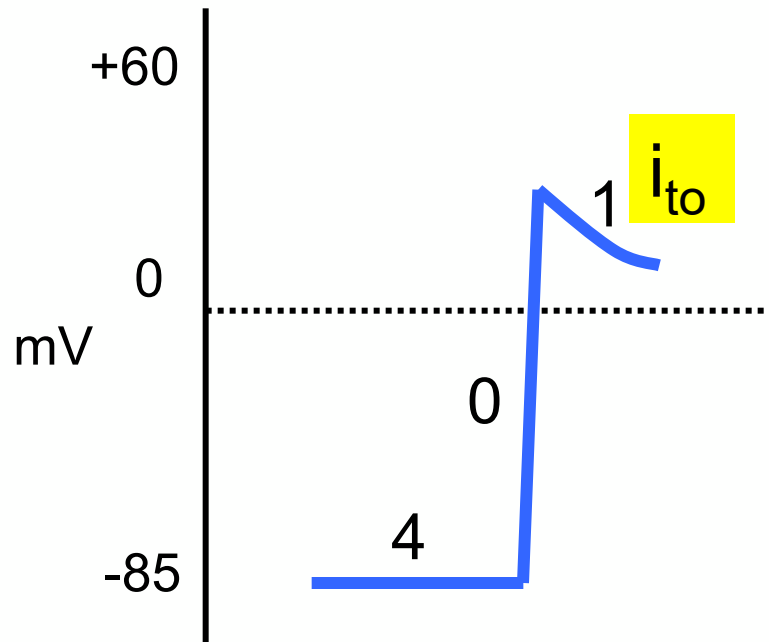
Cardiac Action Potential from a Ventricular Myocyte Cell

Phase 1

Initial Rapid Repolarization

due to

Na^+ inactivation and
outward flow of K^+ (i_{to})



Meanwhile, the voltage-gated L-type Ca^{2+} (L= long lasting) channels have finally opened in response to the depolarization of phase 0. (Start to open at ~ -40 mV, can be phosphorylated β -agonists to $\uparrow$ inotropy)

Inward Ca^{2+} current

Cardiac Action Potential from a Ventricular Myocyte Cell

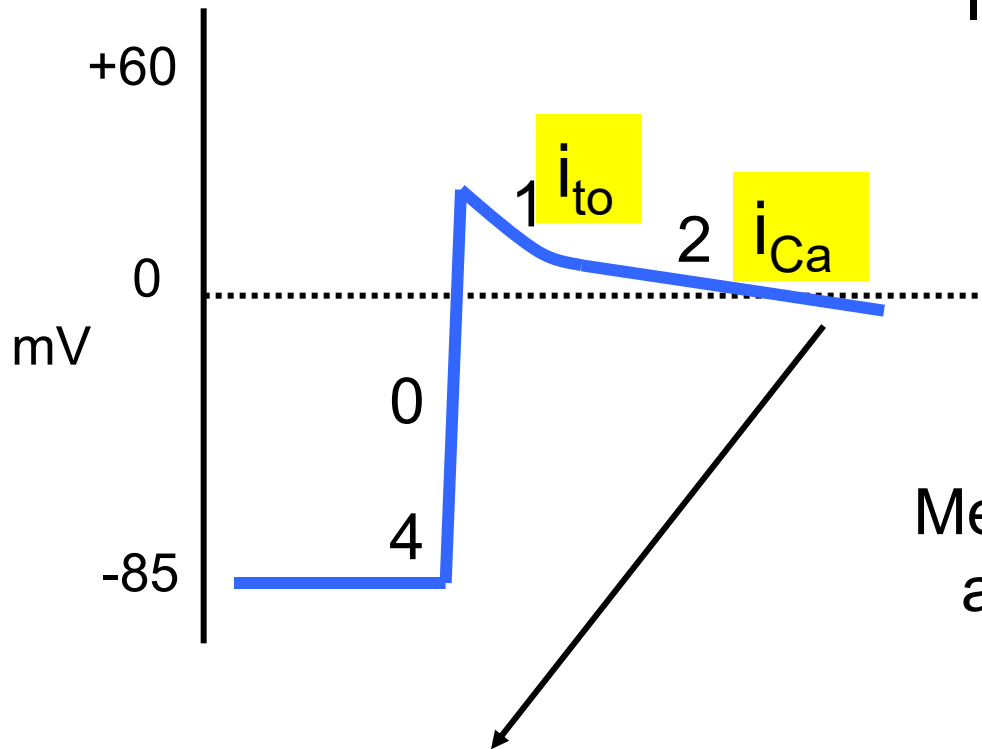
Phase 2

Plateau Phase

Inward flow of Ca^{2+} i_{Ca}

Inward flow of Ca^{2+} slows down the repolarization due to the outward K current (I_{to})

Membrane Potential is “held” at ~ 0 mV for a prolonged period (~ 150 ms).



Ca^{2+} channels start to close and another set of K^{+} channels (delayed rectifiers) start to open

K^{+} flow out I_{K}



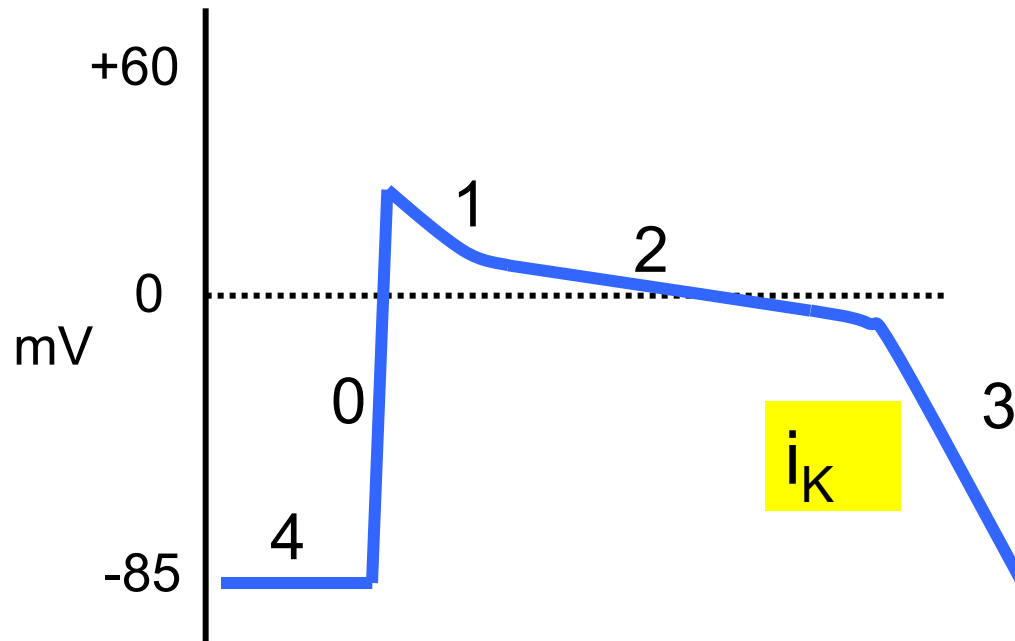
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Cardiac Action Potential from a Ventricular Myocyte Cell

Phase 3

Rapid Repolarization



due to
outward K^+ current I_K
through the delayed
rectifier channels.
slow repolarization
– not propagated
– several potassium channels

Membrane
potential returns
to the resting
level



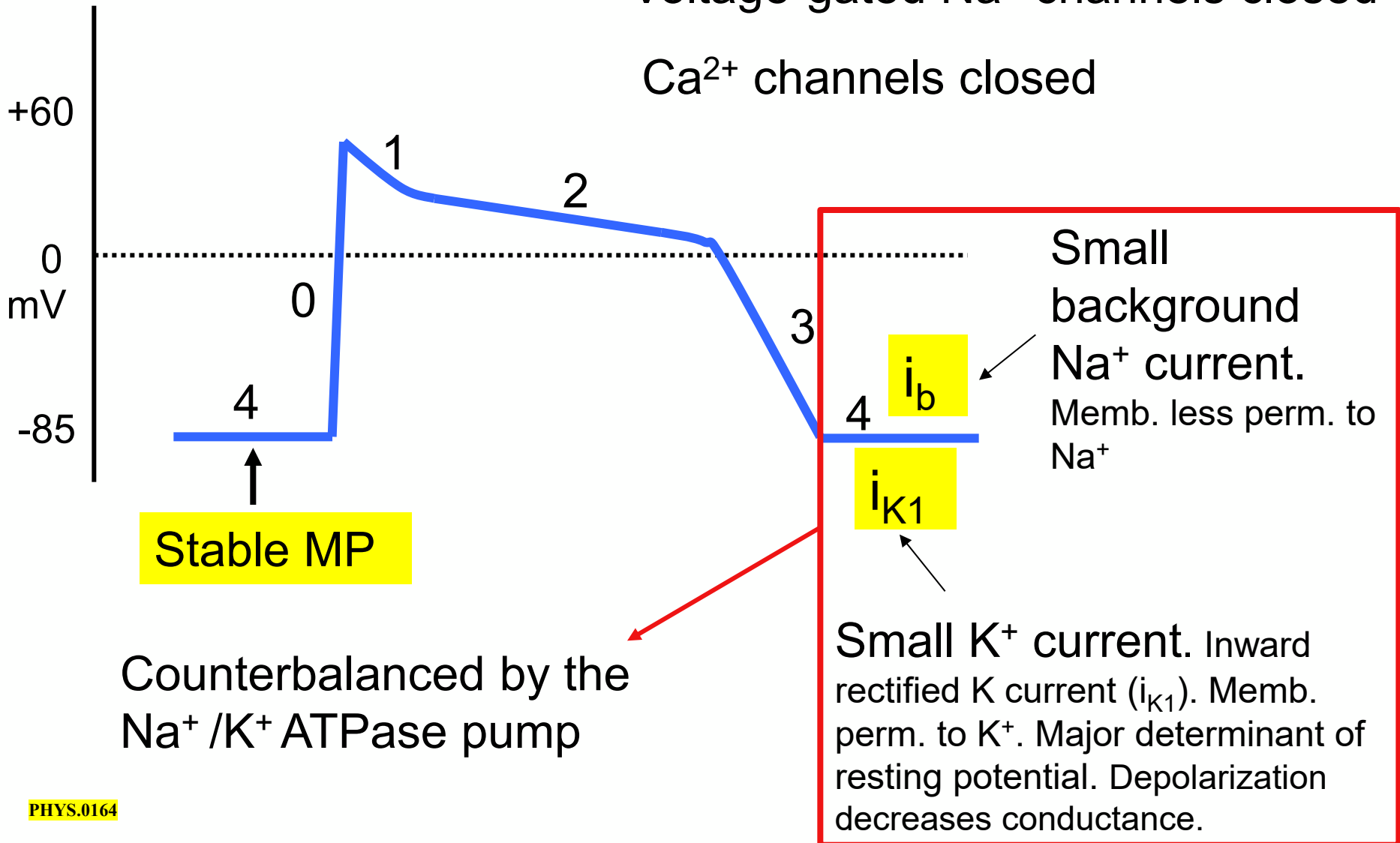
Cardiac Action Potential from a Ventricular Myocyte Cell

Phase 4

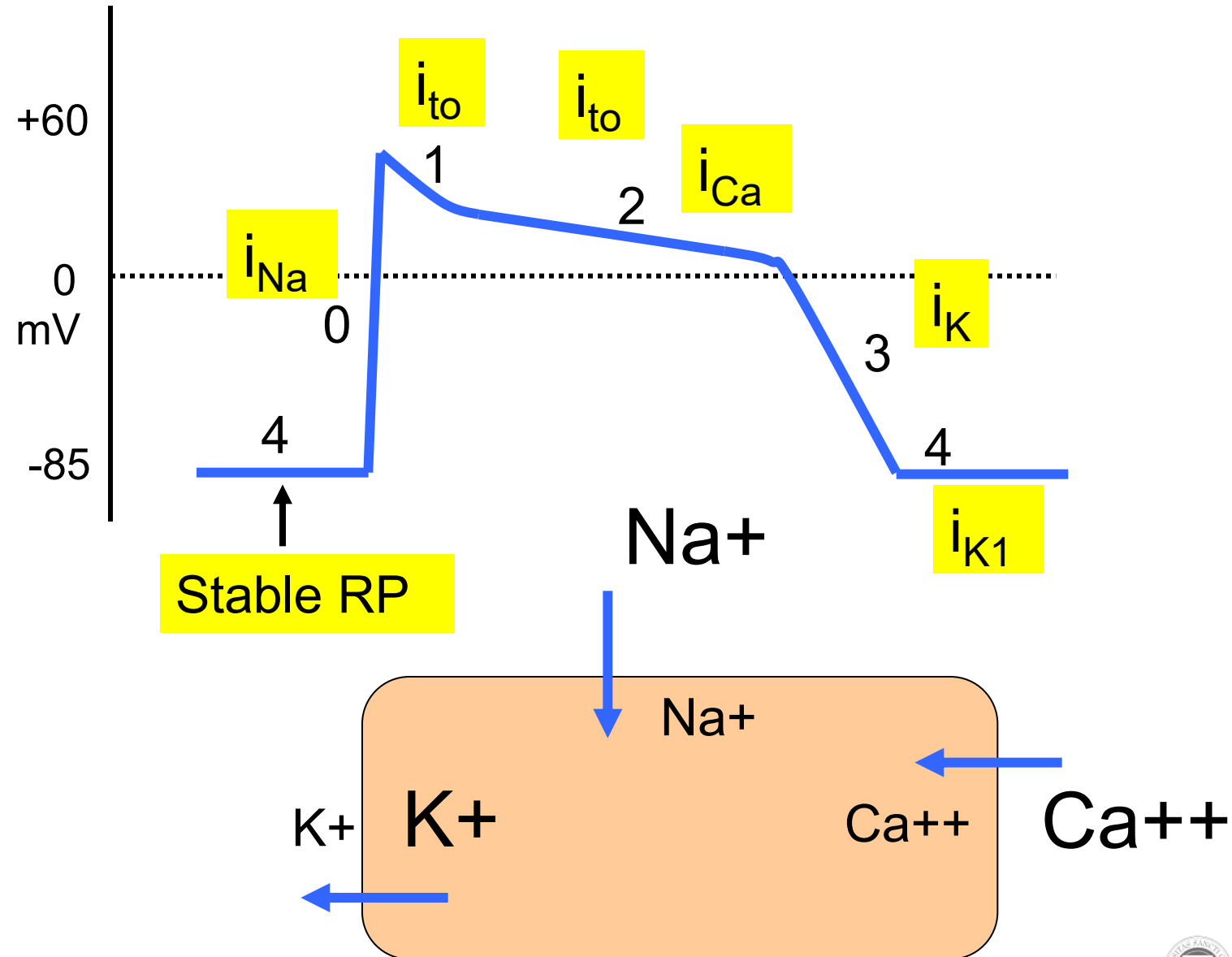
Resting Phase

Voltage-gated Na^+ channels closed

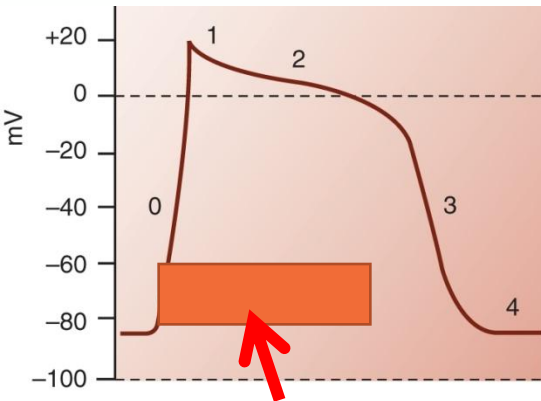
Ca^{2+} channels closed



Cardiac Action Potential from a Ventricular Myocyte Cell



Cardiac Action Potential- cationic currents



Action Potential due to:
Electrochemical gradient
Conductance (which channels are open)

Long depolarization mainly due to Ca^{2+}



i_{Na}

Channel conductance

Down = inward current

Upwards = outward current

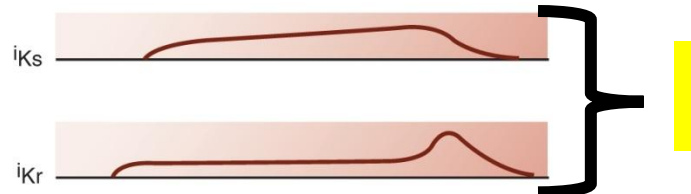


i_{Ca}



i_{to}

Transient outwards (slow, fast)



i_{K}

Delayed rectifiers (slow, fast)



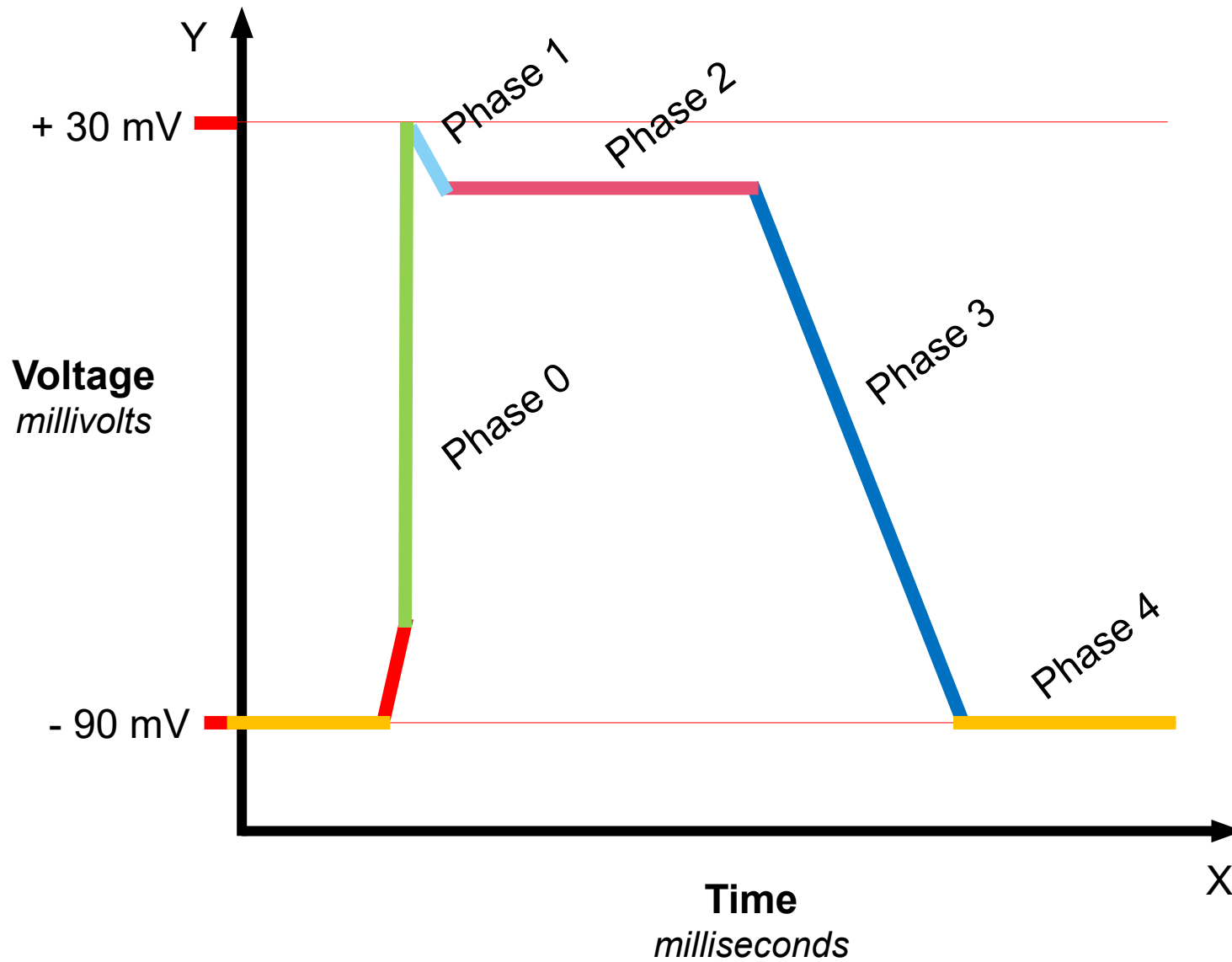
i_{K1}

Resting membrane potential

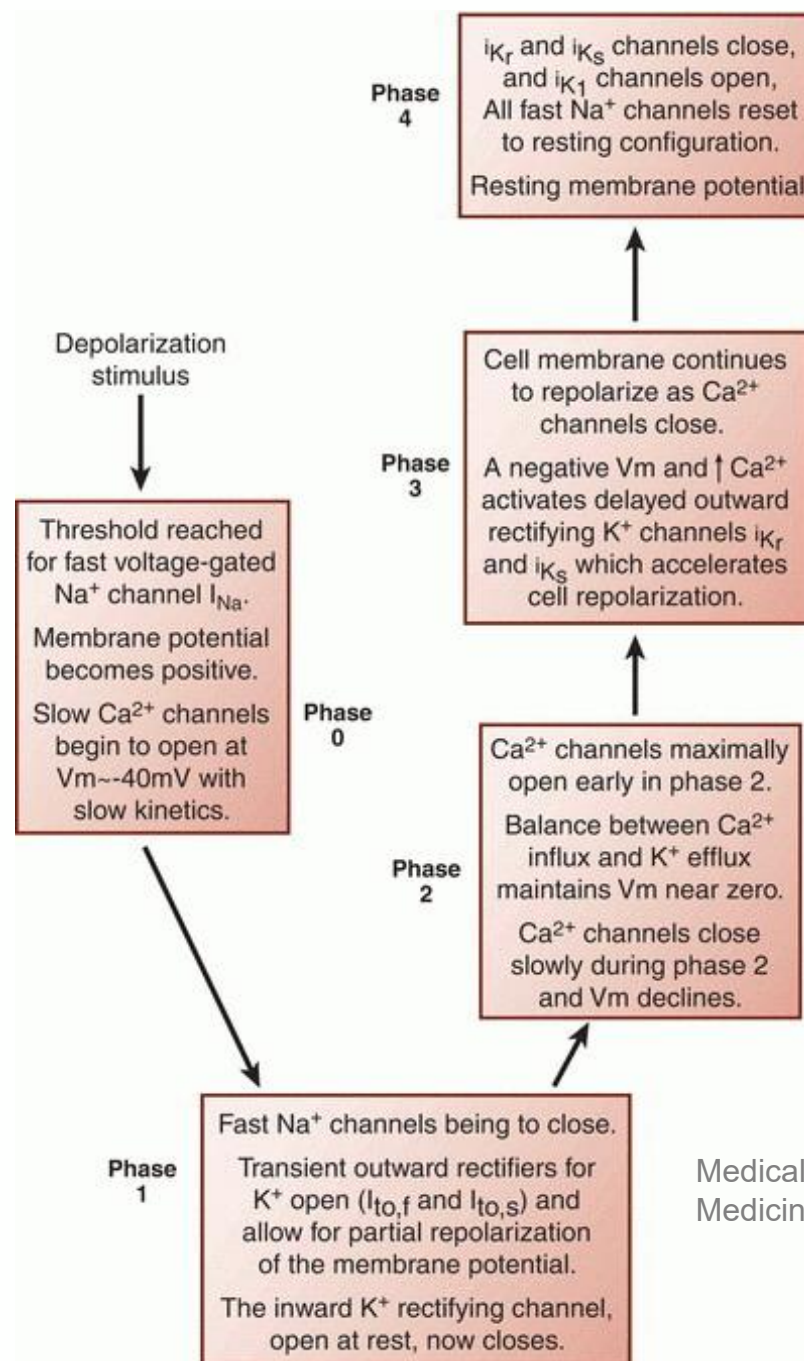


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Ventricular Myocyte Action Potential



Medical Physiology: Principles for Clinical Medicine, 5e, 2017

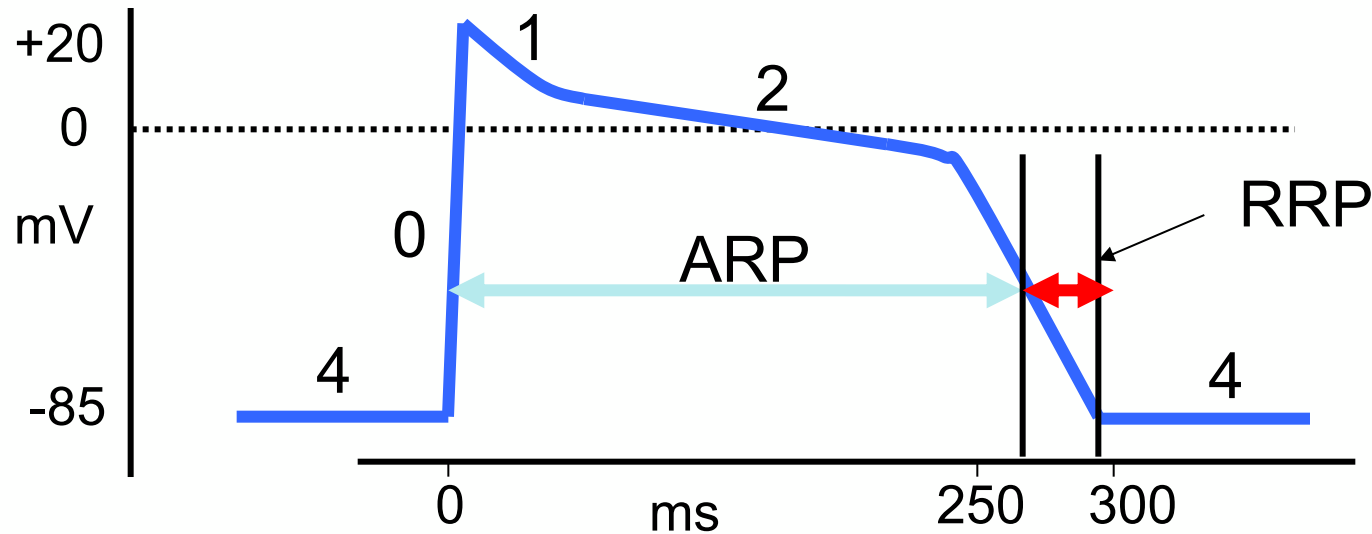
REFRACTORY PERIOD



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Refractory Period of the Cardiac Action Potential

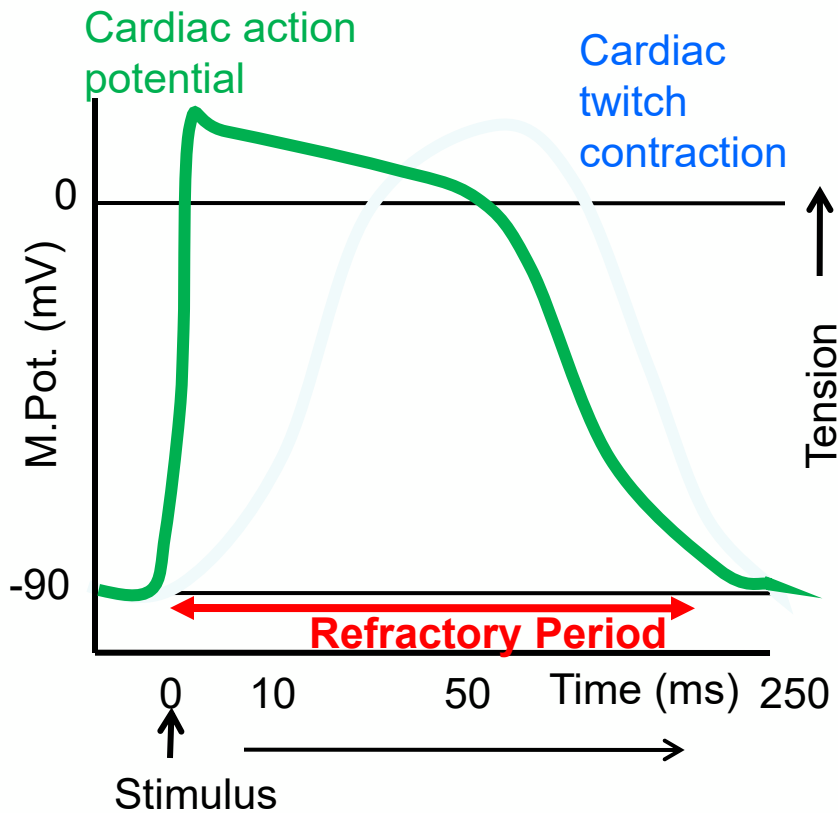


Absolute refractory period (ARP)- Depolarization block of voltage gated Na^+ channels. Requires repolarization for inactivated h-gate to be reset (*Hodgkin-Huxley model- m and h gates*).

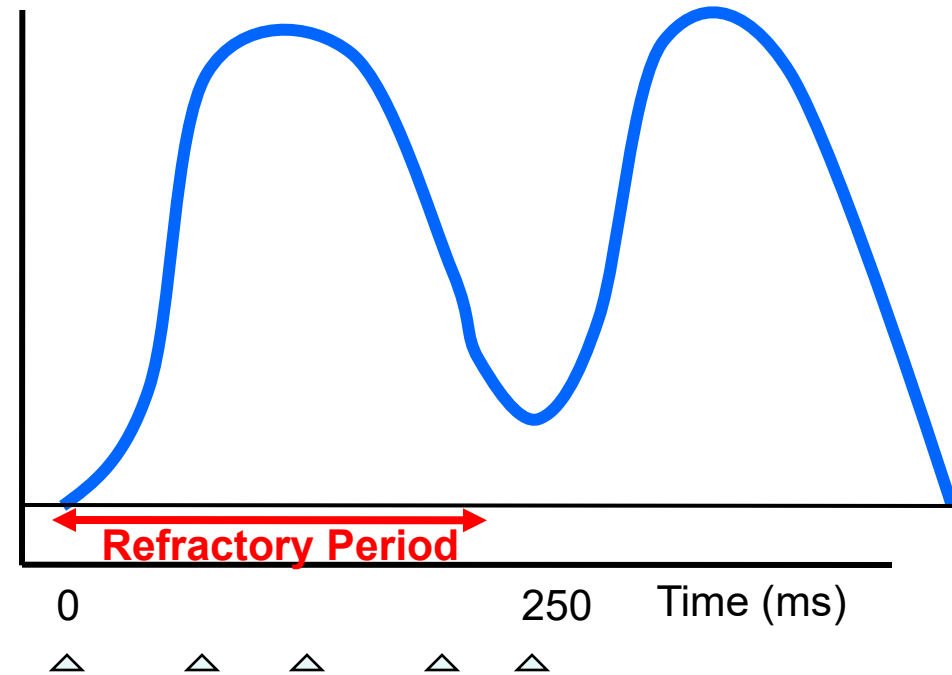
- No stimulus of any magnitude evokes propagated response (infinite threshold).

Relative Refractory Period (RRP)- stimulus above normal threshold generates a response (re-entry circuits-pathology).





Long Refractory Period in Cardiac Muscle prevents tetanic contractions



Advantage- no tetanus. Heart cannot fill if in continuous and max contracted state.
 Disadvantage- Ectopic currents-heart beat arises from fibers outside SA node. Can render the heart refractory to normal signal from SA node.



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Absolute Refractory Period **ARP**

During the phases 0, 1, 2, and part of phase 3, the cell is refractory (unexcitable)

During the **ARP**, stimulation of the cell does not produce new, propagated action potentials

ARP acts as a protective mechanism in the heart by limiting the frequency of action potentials and therefore contractions that the heart can generate, enabling the heart to have adequate times to fill and eject blood

The long **ARP** also prevents the heart from developing sustained, tetanic contractions like those that occur in the skeletal muscle



Relative Refractory Period **RRP**

At the end of the **ARP** the cell is in its **RRP**

Early in this period suprathreshold depolarization stimuli are required to elicit action potentials

Because not all the sodium channels have recovered to their resting state by this time, action potentials generated during **RRP** have a decreased phase 0 slope and lower amplitude



What accounts for the long refractory period of the cardiac action potential?

This sustained sodium channel inactivation, combined with activation of calcium channels and the delay in opening of potassium channels, accounts for the long plateau phase and the long cardiac refractory period, which lasts until the end of phase 3 (repolarization)



Antiarrhythmic Class I (SODIUM CHANNEL BLOCKERS)

During the phases 0, 1, 2, and part of phase 3, the cell is refractory (unexcitable)

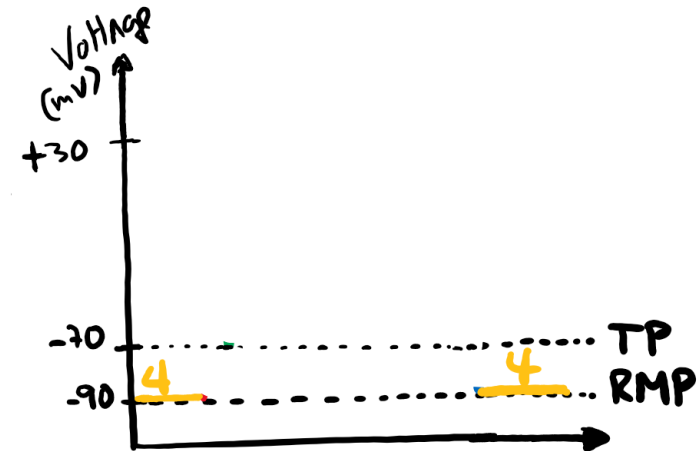
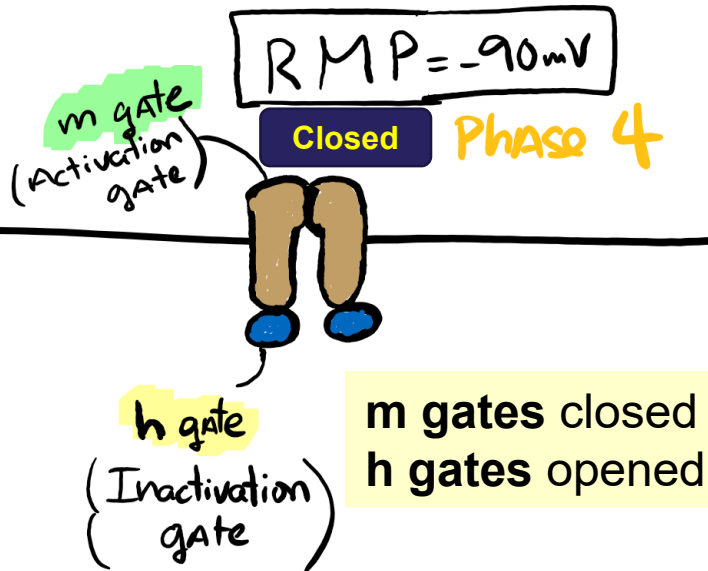
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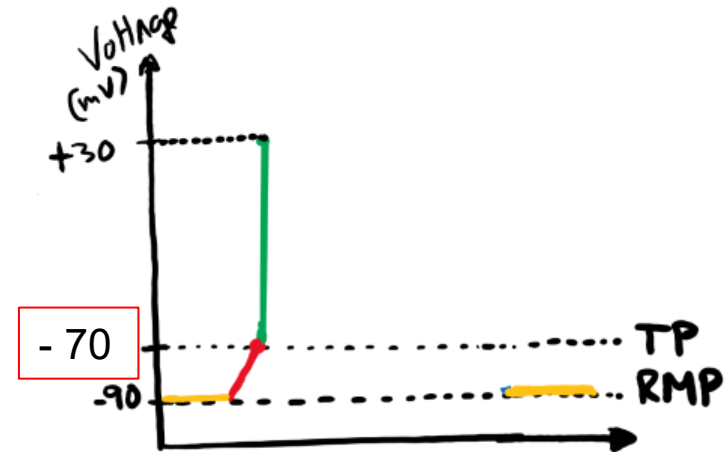
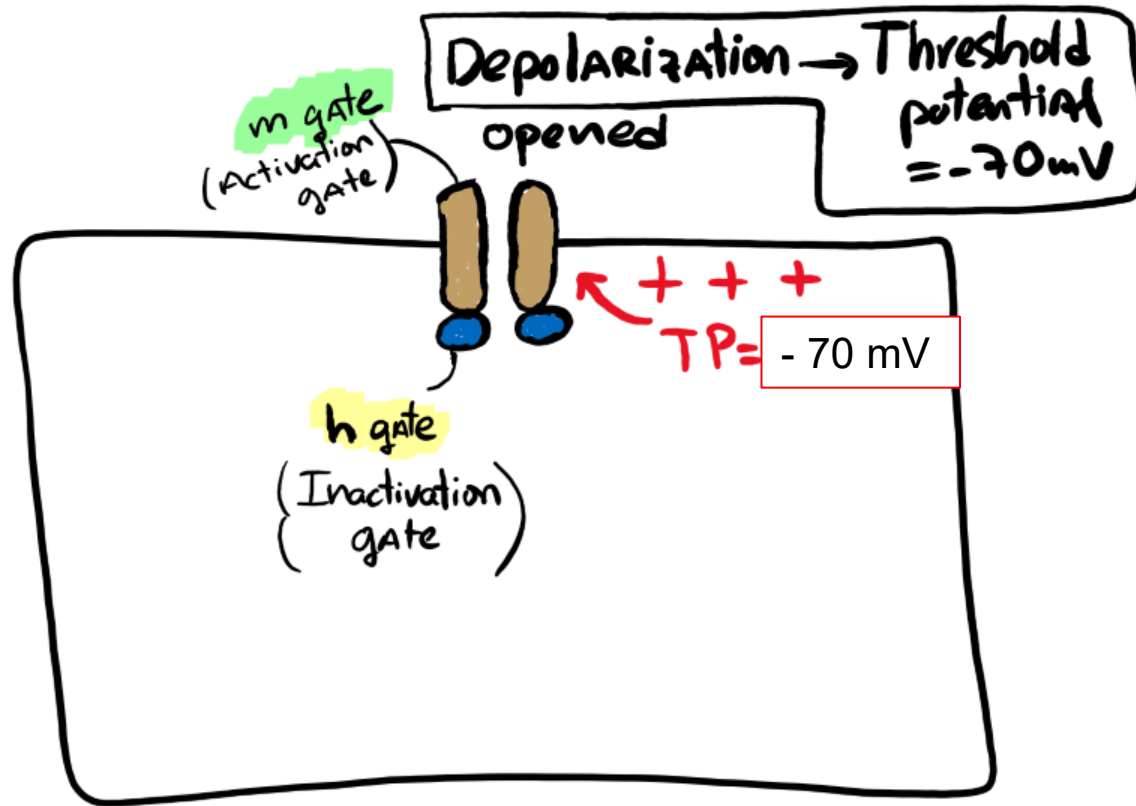
The long **ARP** also prevents the heart from developing sustained, tetanic contractions like those that occur in the skeletal muscle



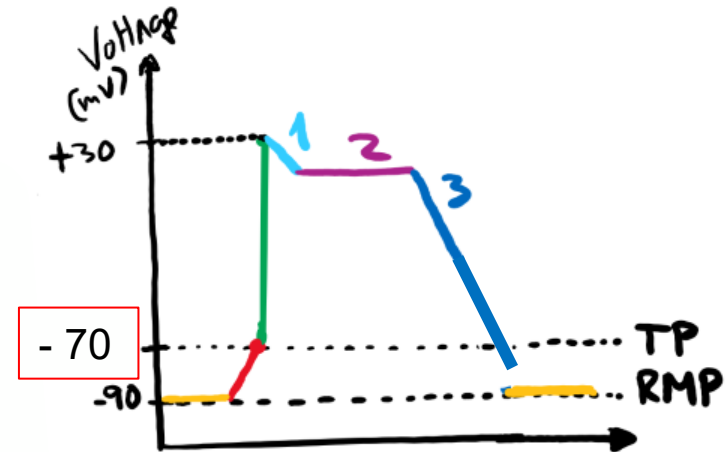
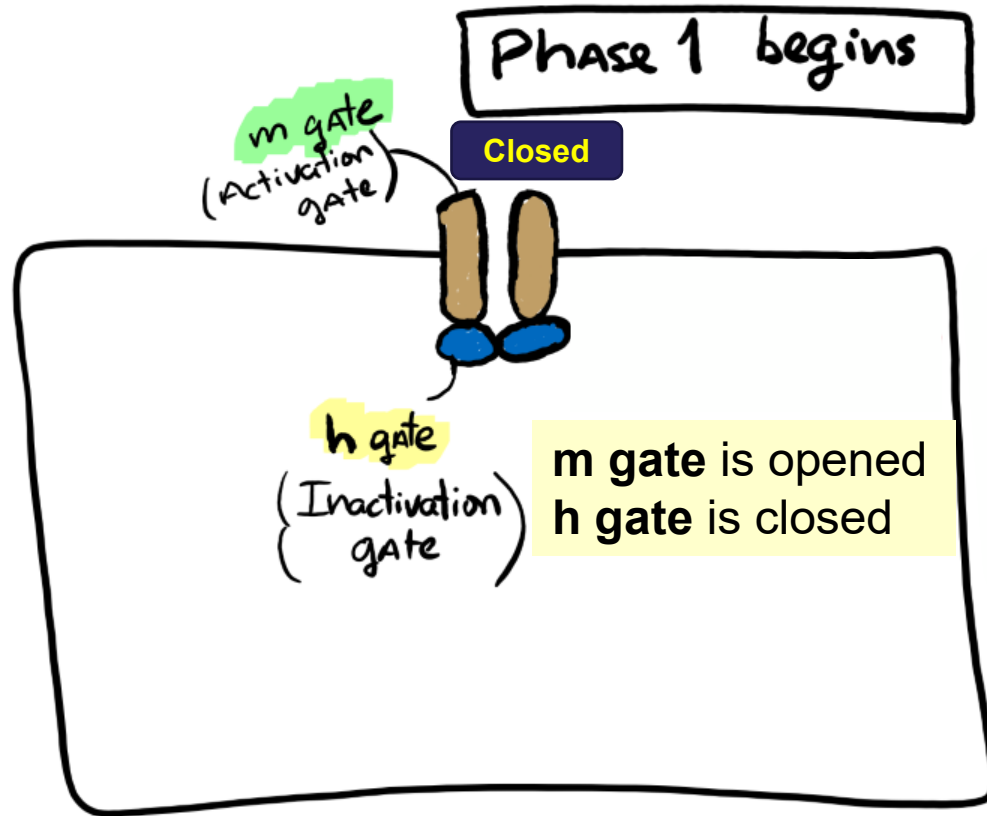
Voltage-gated Sodium Channel



Voltage-gated Sodium Channel



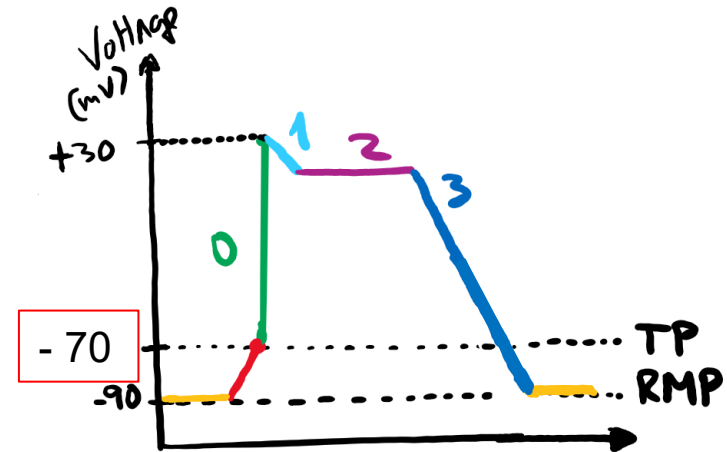
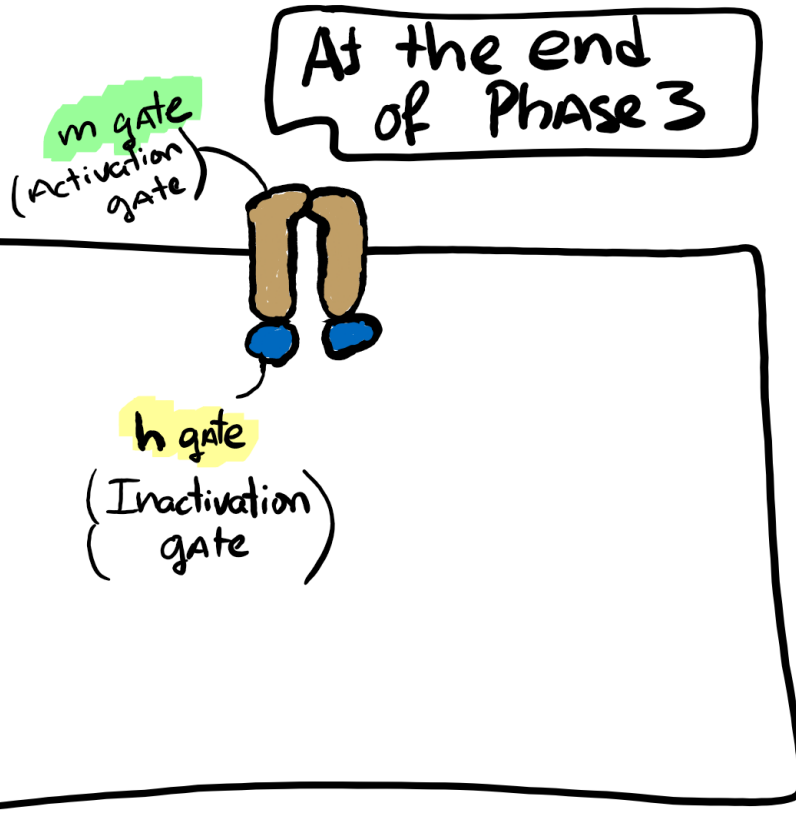
Voltage-gated Sodium Channel



Phase 1
Phase 2
part of Phase 3



Voltage-gated Sodium Channel



Conducting System of the Heart

Progression of Wave of Depolarization through the Heart

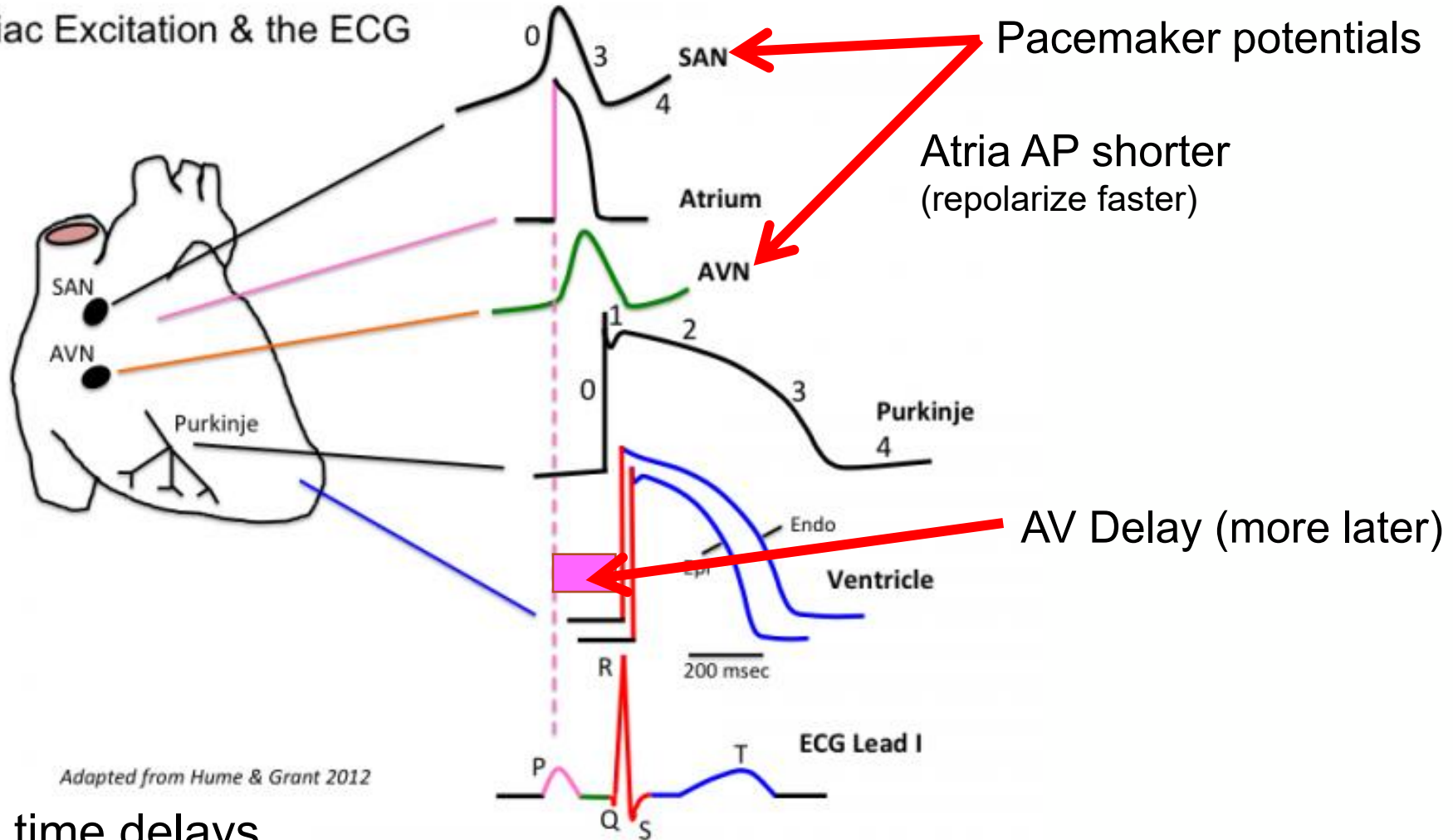


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Putting all Action Potentials of heart together

Cardiac Excitation & the ECG



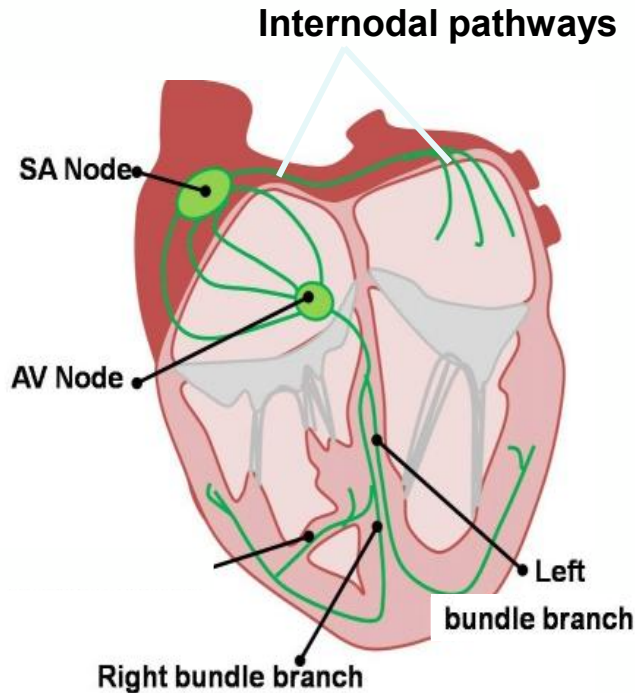
Adapted from Hume & Grant 2012

Note time delays

SA before AV (sets pace; overdrive suppression-hyperpolarization of other pacemakers)

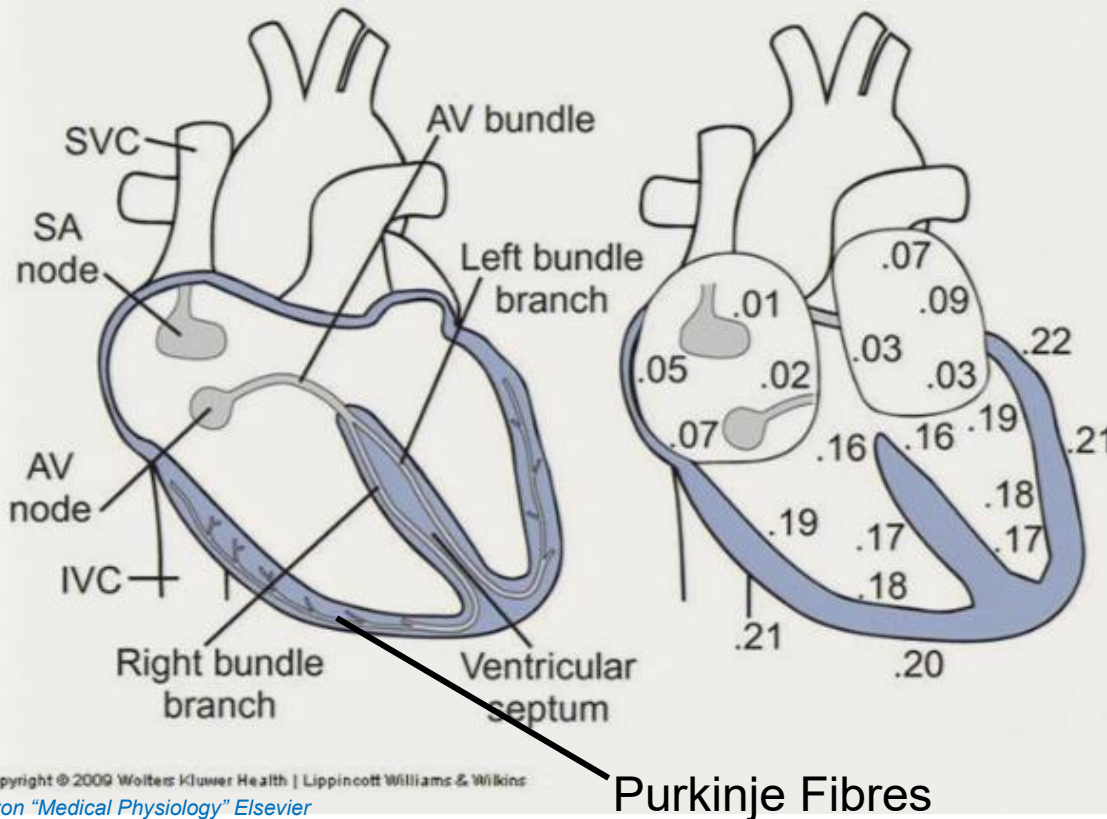
Delay between Atrial and Ventricular contraction- due to AV node

Conducting System of the Heart



- Heart rate initiated by SA node. (ANS modulate HR).
- Depolarization of myocytes spread via gap junctions- syncytium
- Special system for rapid conduction of depolarization through heart (especially ventricles)- conducting system.
- Advantages of conducting system:
 - Control timing of contraction
 - Control which regions contract

Conducting System of the Heart -ordered contraction



SA node



Internodal branches



AV node



(a major site of ANS regulation)

Bundle of His



Purkinje fibres

Ordered contraction of heart

- Atria first- allowing time to fill ventricle
- Ventricles- contract septum, then apex to base. (efficient ejection of blood)

Speed of conduction:

Bundle branches=Purkinje~AV bundle

> atrial myocard. > ventricle myocard. > SA node > AV node

OR 1. electrical conduction system > 2. myocardium > 3. nodes



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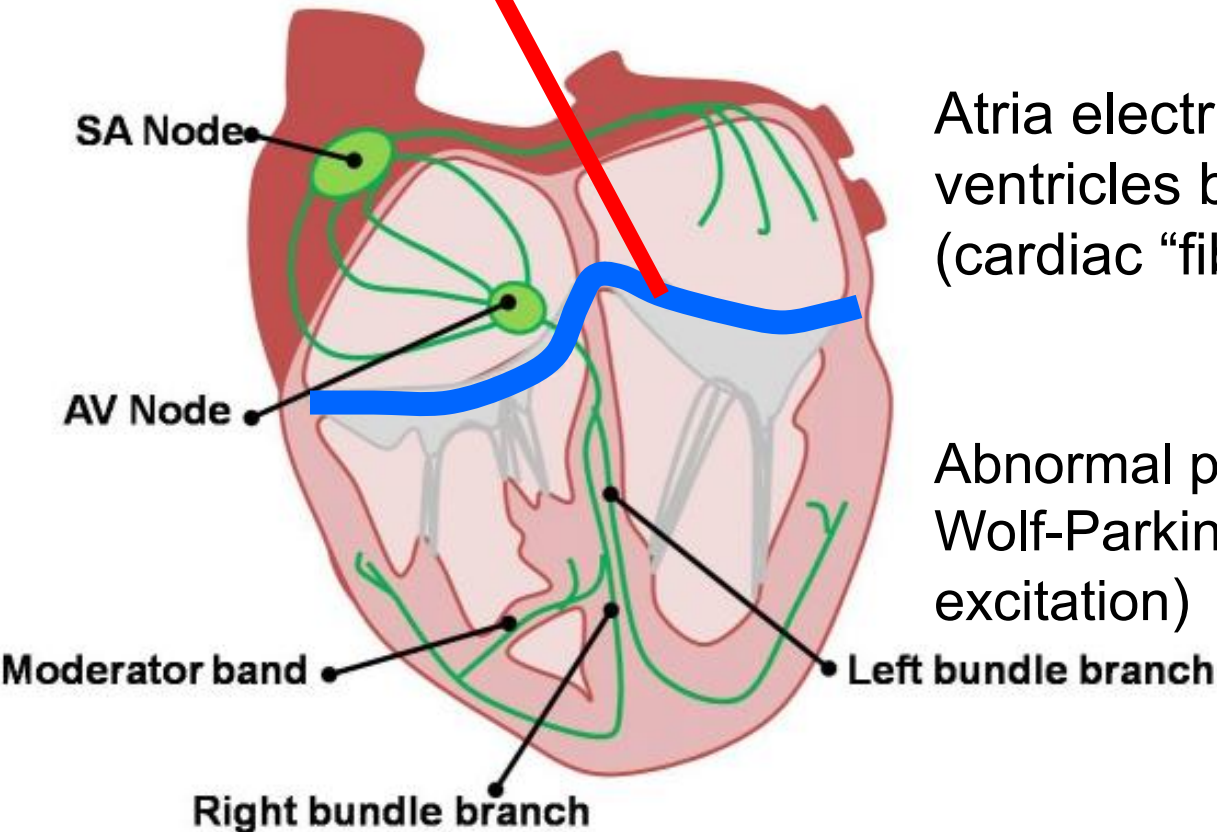
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Conducting System of the Heart

Fibro-adipose tissue
“insulator”

**Only normal path for
depolarization is through-
atrioventricular (AV) node**

Atria electrically isolated from
ventricles by fibro-adipose tissue
(cardiac “fibrous skeleton”)



Abnormal paths- Bundle of Kent →
Wolf-Parkinson-White syndrome (pre-
excitation)



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Conducting System of the Heart- second importance

Conduction DELAYED at AV Node

Advantages

Allows time for atria to contract before ventricles contract

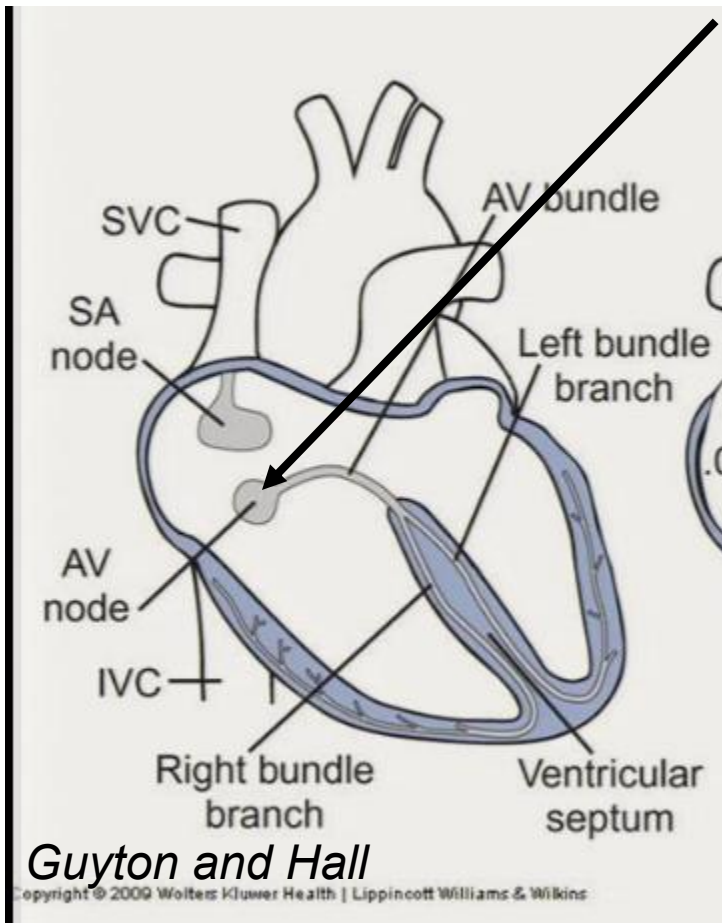
– atria- topping up ventricles (~10% at rest, 40% during exercise).

Atrial tachyarrhythmias may not be transmitted to the ventricles – safeguarding the ventricles.

Disadvantages of delay

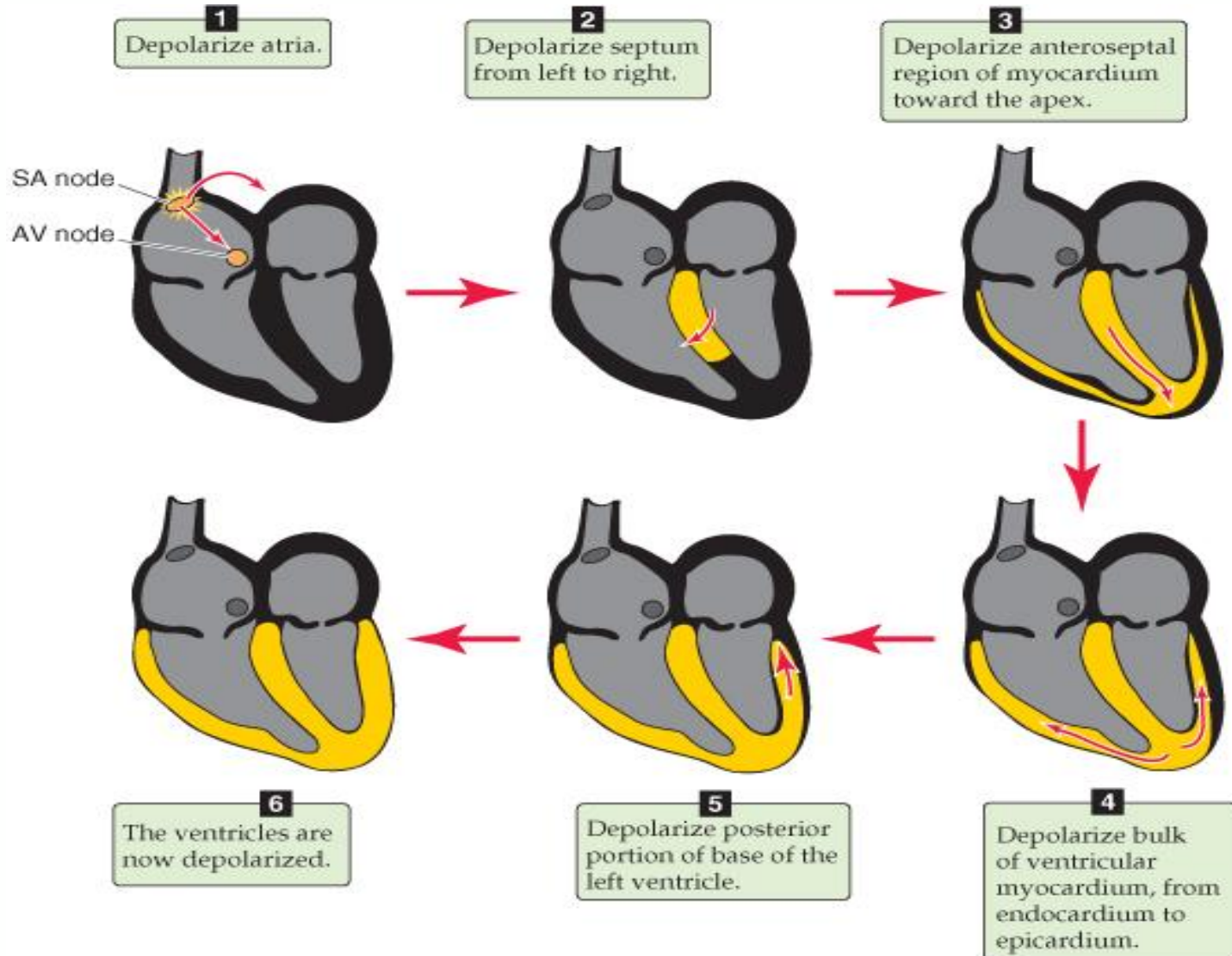
Low safety factor for transmission

-discuss pathologies of conduction system later



AV node – ONLY normal electrical contact between atria and ventricles

Progression of Wave of Depolarization through the Heart



Conducting System of the Heart

SA node: 60 – 75 bpm
(normal sinus rhythm)

AV node: 50 bpm

Bundle of His: 40 bpm

Purkinje fibres: 15 bpm

However, HR is set by the fastest – the SA node.

So AV cannot drive HR under normal conditions.

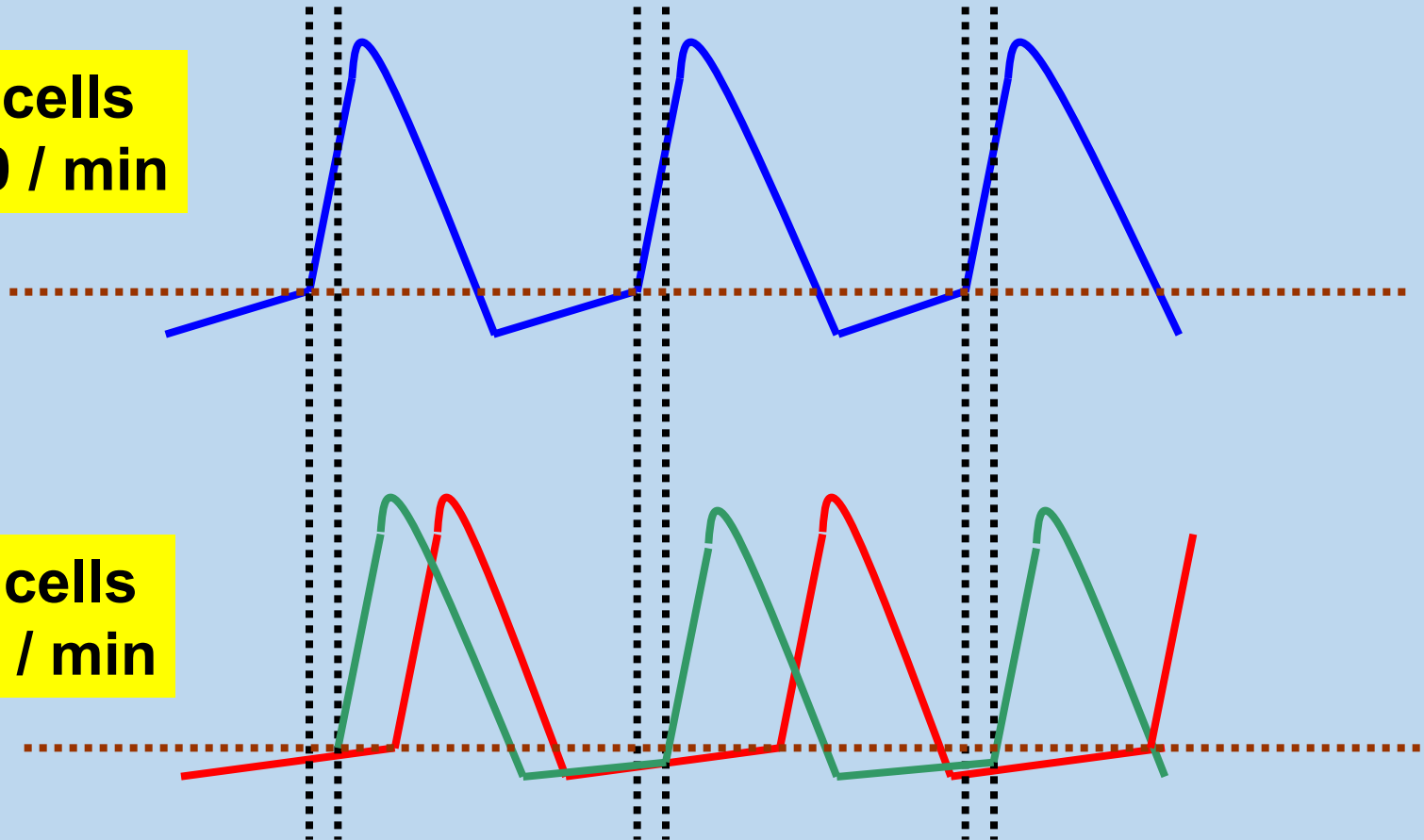
- SA node primary pacemaker → overdrive suppression. Leads to hyperpolarization of other pacemakers sites.



Conducting System of the Heart- SA node drives heart rate

SAN cells
60-100 / min

AVN cells
50-60 / min



- SA node primary pacemaker (**blue line**)
- AVN threshold = longer time to depolarization (**red line**)
 - Pacemaker potential reset (**green line**) on depolarization by SA node



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Sinoatrial (SA) node Action Potential

Pacemaker potential



Location of Pacemaker Cells

Sinoatrial (SA) Node

Located within the posterior wall of the right atrium

Primary pacemaker site of the heart

Atrioventricular (AV) Node

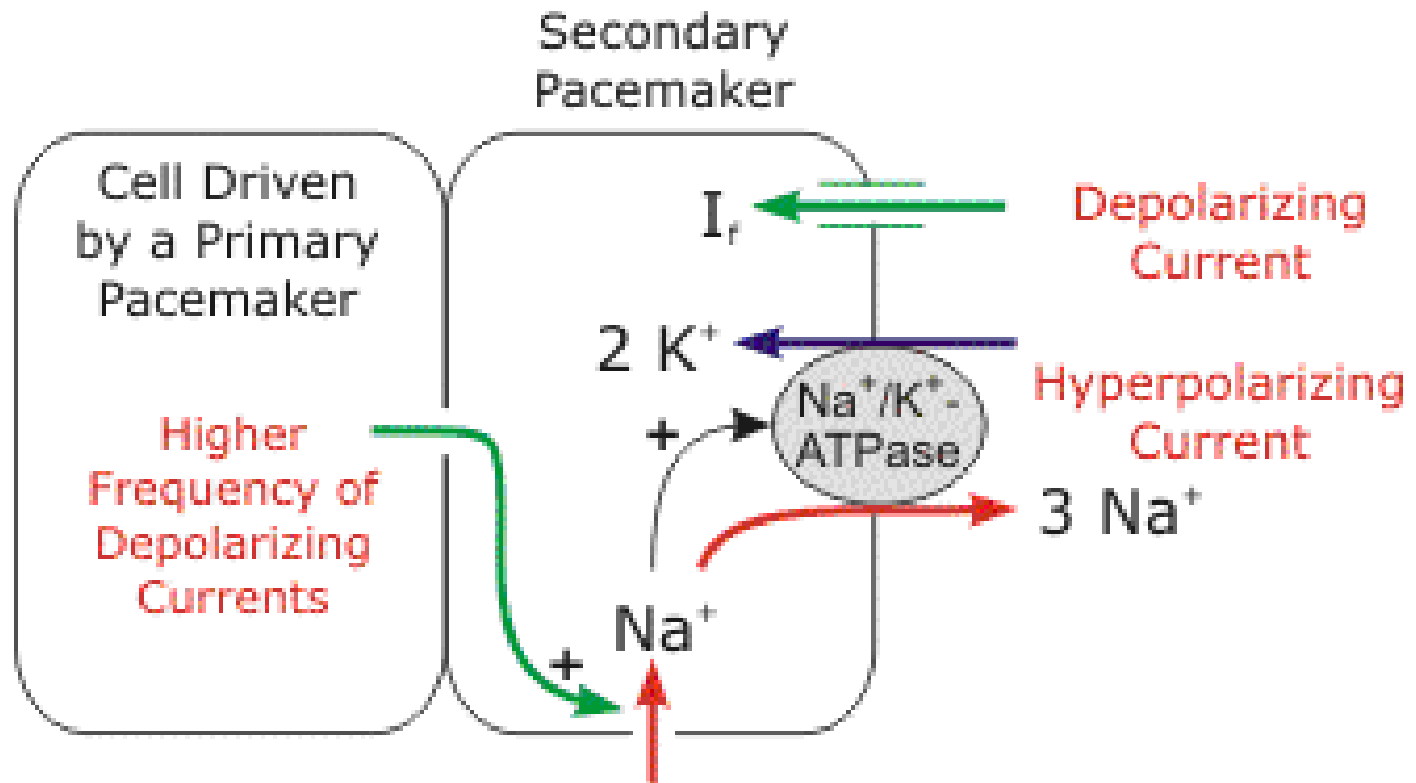
Located at the base of the right atrium within the triangle of Koch, a region defined by the following landmarks: the coronary sinus ostium, tendon of Todaro, and the septal leaflet of the tricuspid valve.

Other pacemaker cells exist within the ventricular conduction system

Why the firing rate of all the pacemaker cells are driven by the higher rate of Sinoatrial (SA) Node?

Overdrive suppression

Location of Pacemaker Cells



Overdrive Suppression

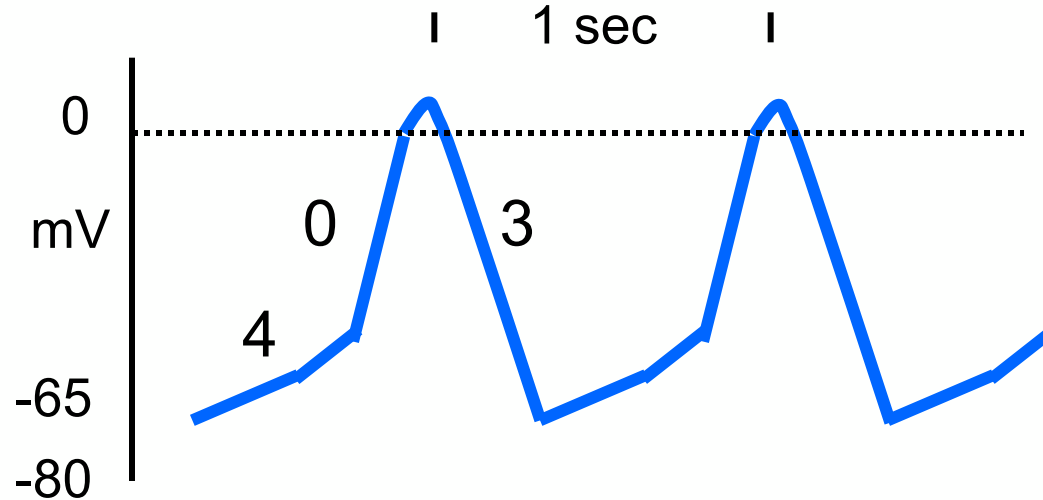
This mechanism causes the secondary pacemaker to become hyperpolarized when driven at a rate above its intrinsic rate

Hyperpolarization occurs because the increased action potential frequency stimulates the activity of the electrogenic Na/K ATP-ase pump as a result of enhanced entry of sodium per unit time into these cells

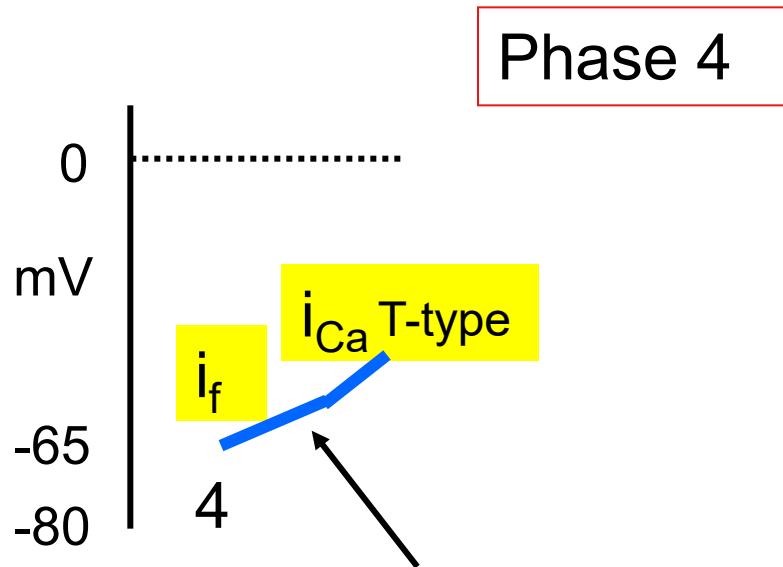
If the SA Node become depressed, or its action potential fail to reach secondary pacemakers, overdrive suppression ceases, which permits a secondary site to take over as the pacemaker for the heart. When this occur, this new pacemaker is called an **ectopic foci**

Action Potential from the sinoatrial (SA) node

Pacemaker potential



Action Potentials from the SA node



Pacemaker Potential
UNSTABLE

Pacemaker Potential

(Slow diastolic potential)

due to

Inward Na^+ current i_f

Inward Ca^{2+} current i_{Ca}

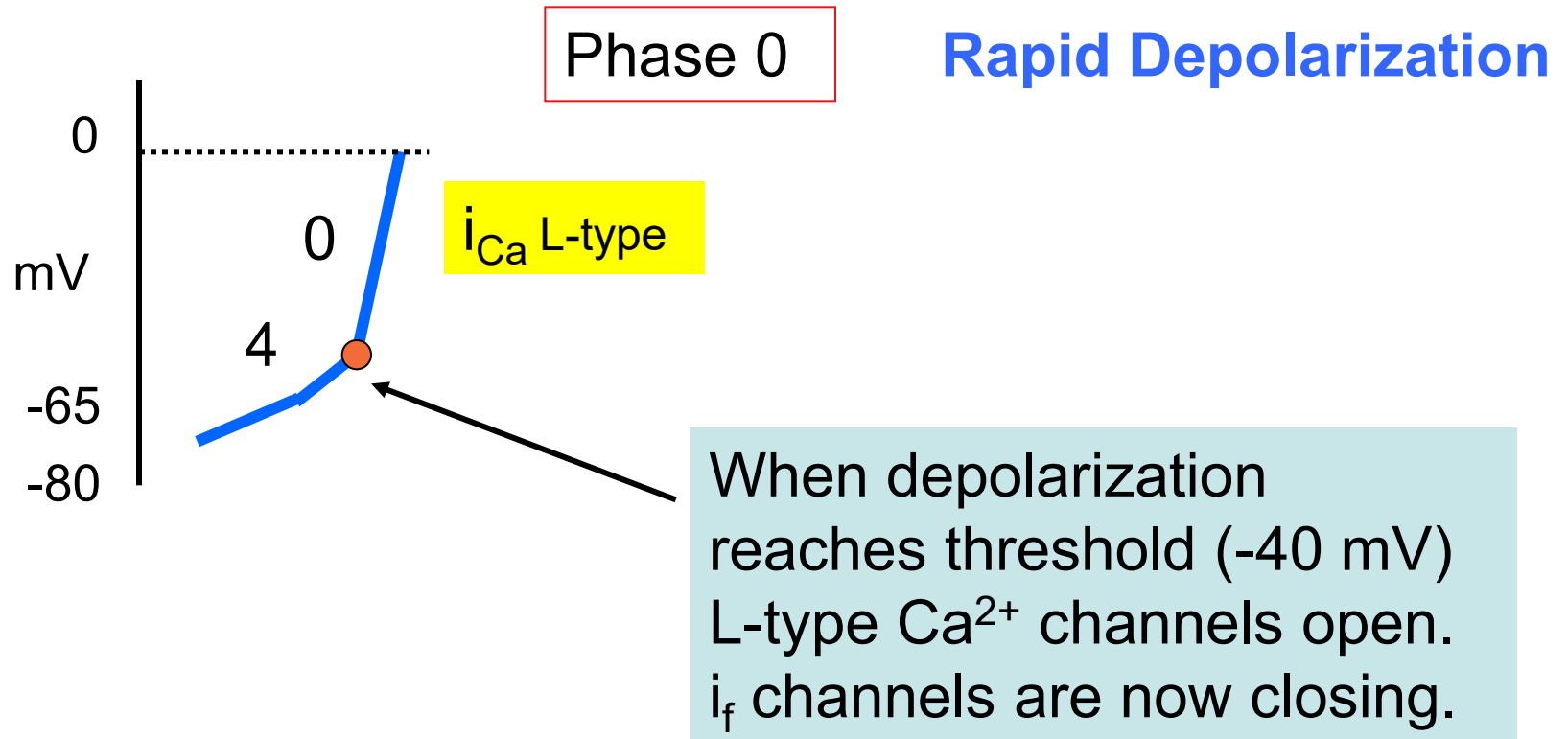
Na channels are “funny” -
they open when
membrane is relatively
polarized or at rest
(unusual !)

i_f channels open $\rightarrow$ inward Na^+ current $\rightarrow$ Depolarization



More depolarization $\leftarrow i_{Ca} \leftarrow$ Open T-type Ca^{2+} channels
T= transient

Action Potential from the SA node



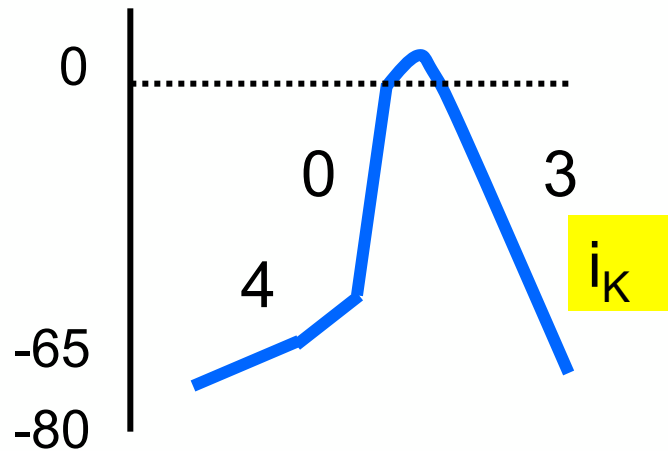
As Ca^{2+} channels start closing, K^+ channels start opening in response to the depolarization



Note: Phase 1 and 2 absent to keep nomenclature similar to cardiac AP

Phase 3

Repolarization



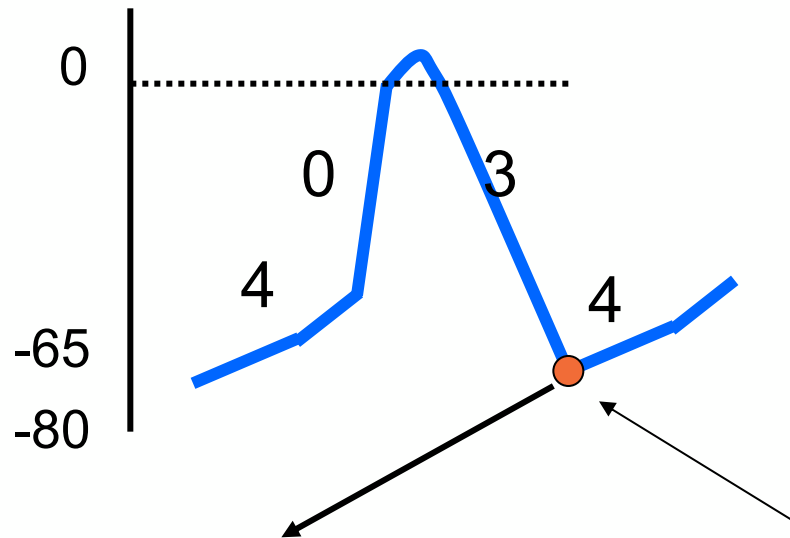
due to
Closure of Ca^{2+} channels

Opening of K^+ channels

As the membrane becomes more
and more repolarized the Na^+ i_f
channels start to open again

Phase 4

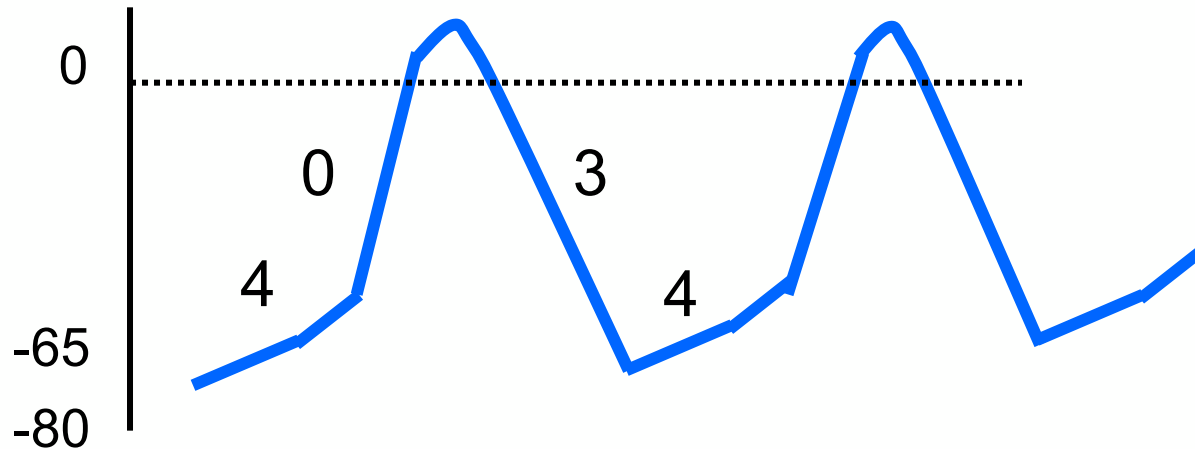
Pacemaker Potential



Repolarization opens the Na⁺ i_f channels

Na⁺ i_f channels open and membrane starts to SPONTANEOUSLY depolarize again

Action Potential from the SA node



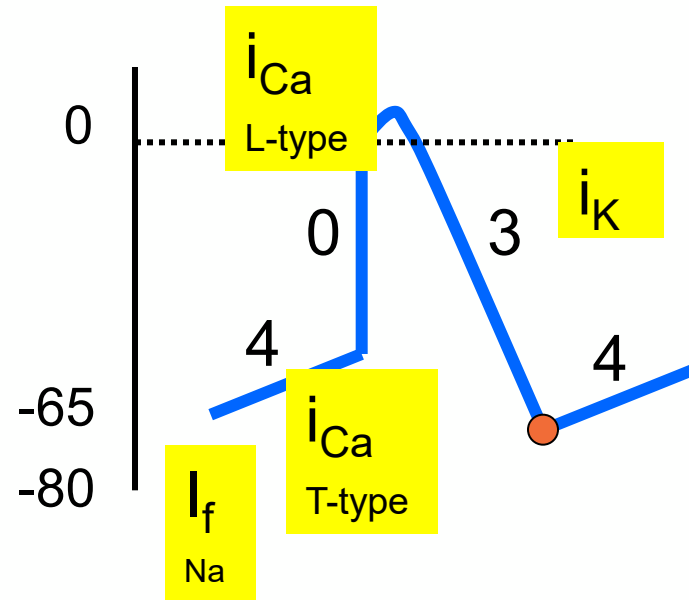
Characteristic Features

Unstable resting potential

No plateau phase

Upstroke of action potential (phase 0) is due to inward Ca^{2+} current (**not Na^{+}**). Upstroke slower than in ventricular AP

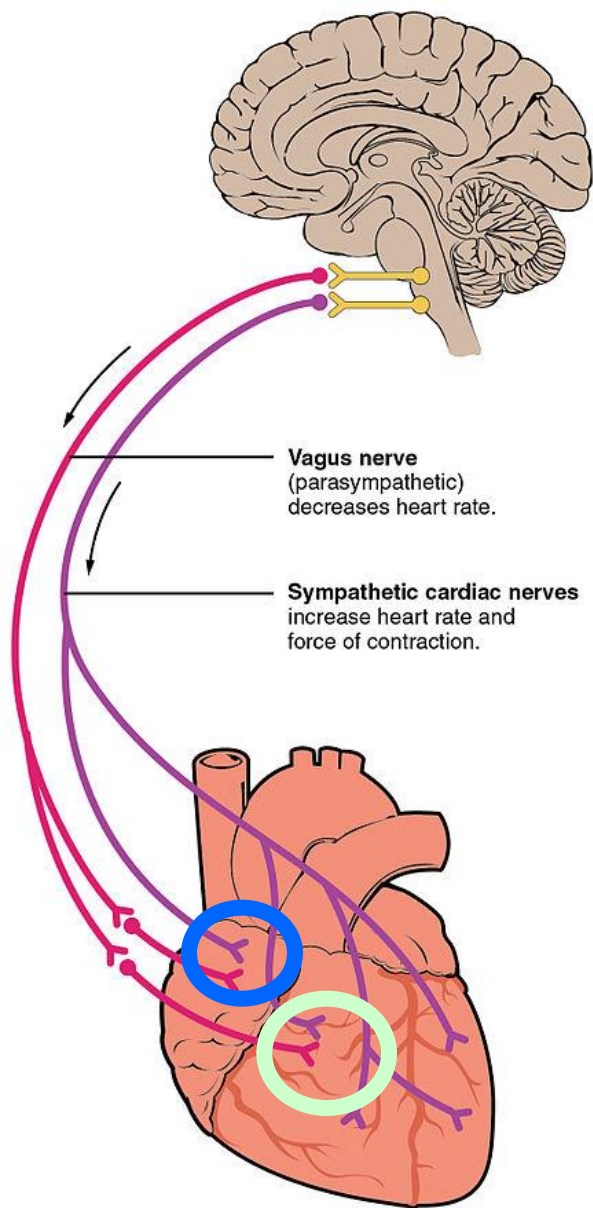
Summary- Action Potential from the SA node



ANS regulation of SA node frequency

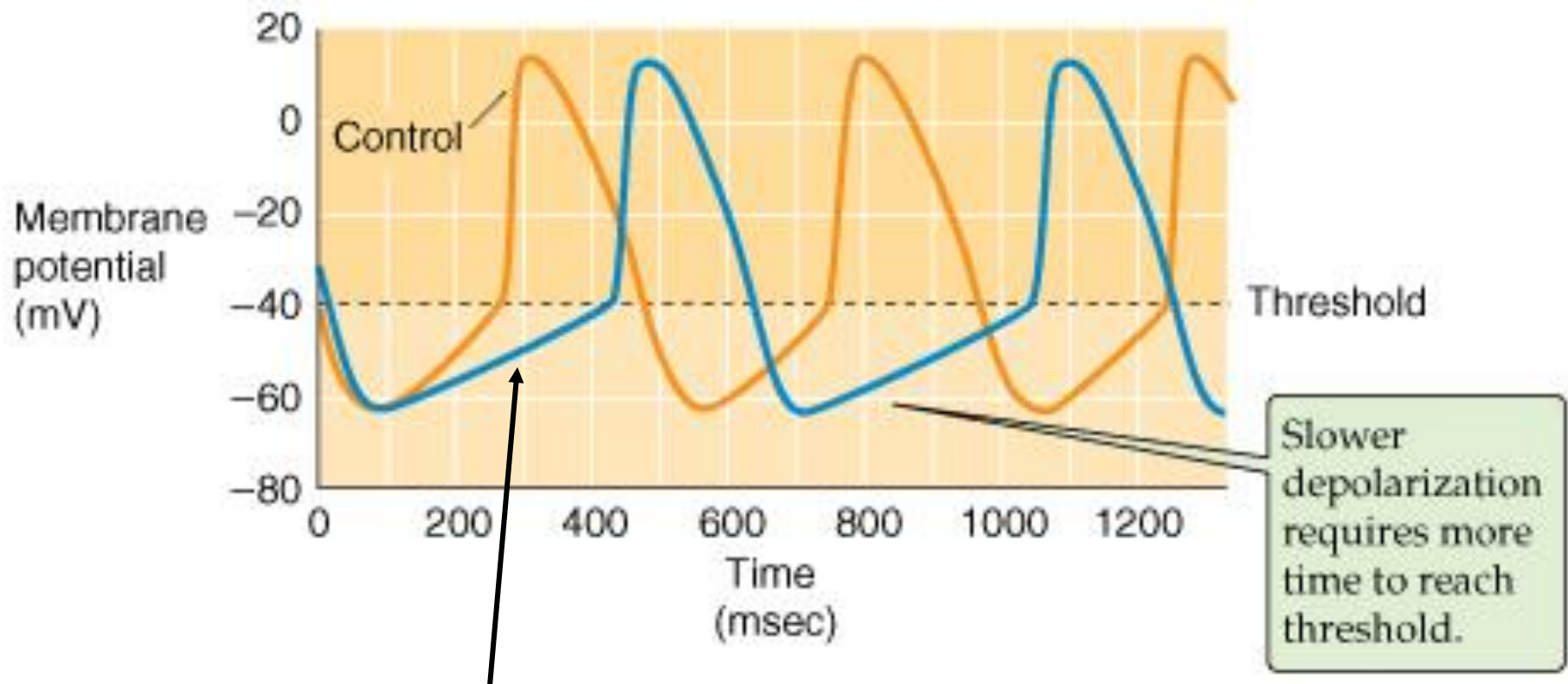
Chronotropy





	Sympathetic	Parasympathetic
SA node	Inc activity: -inc slope phase 4 AP -inc SA node firing -dec vagal tone	Vagal tone dominant resting heart. Inc activity: -hyperpolarize -dec slope phase 4 AP
AV node	Inc conduction velocity	dec conduction velocity
Atrial myocyte	Innervated -inc atrial inotropy	Innervated -dec inotropy
Ventric. myocyte	Little innervation (gap- junctions syncitium)	Little innervation

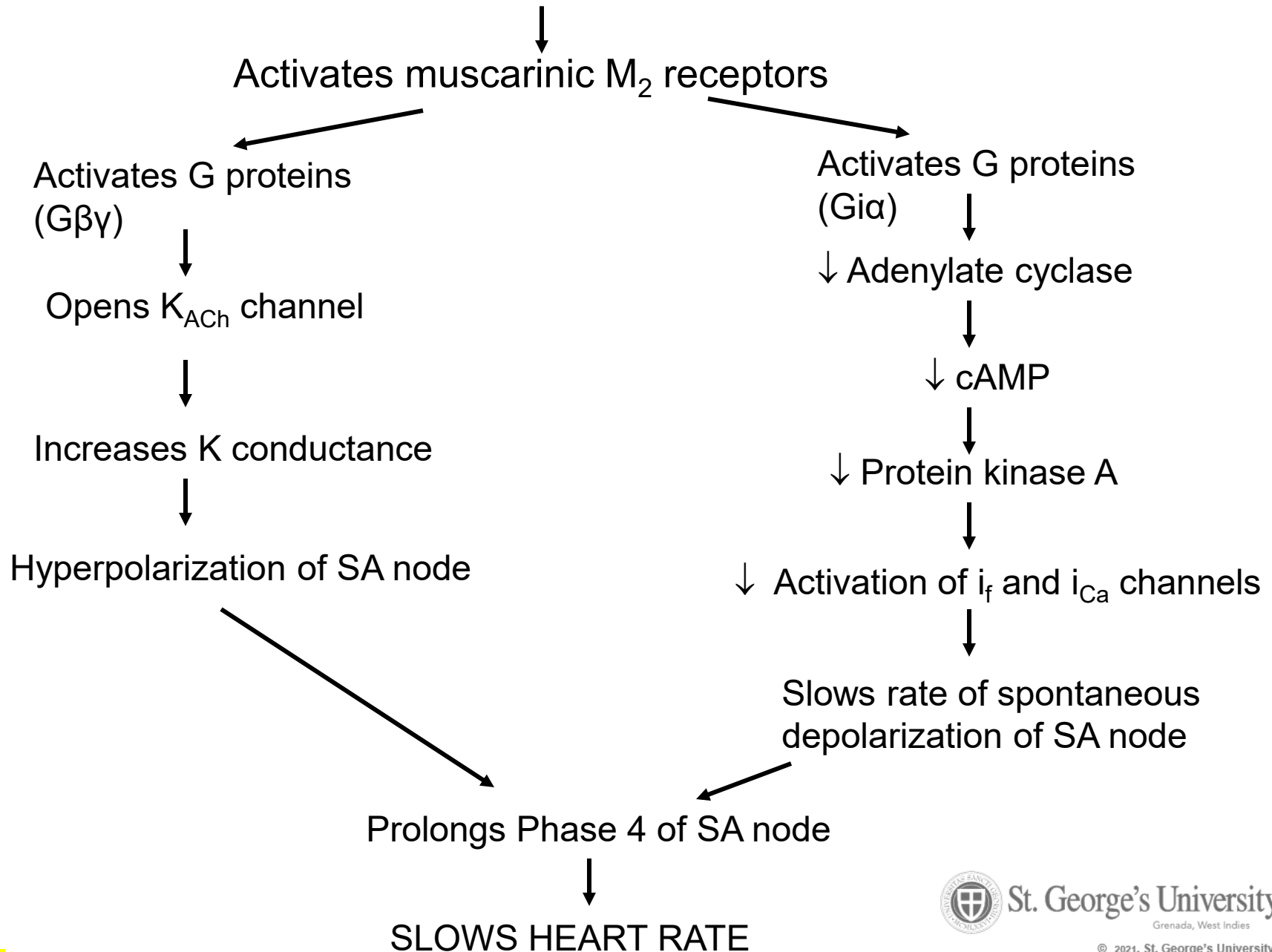
Effects of Parasympathetic Stimulation on the SA node



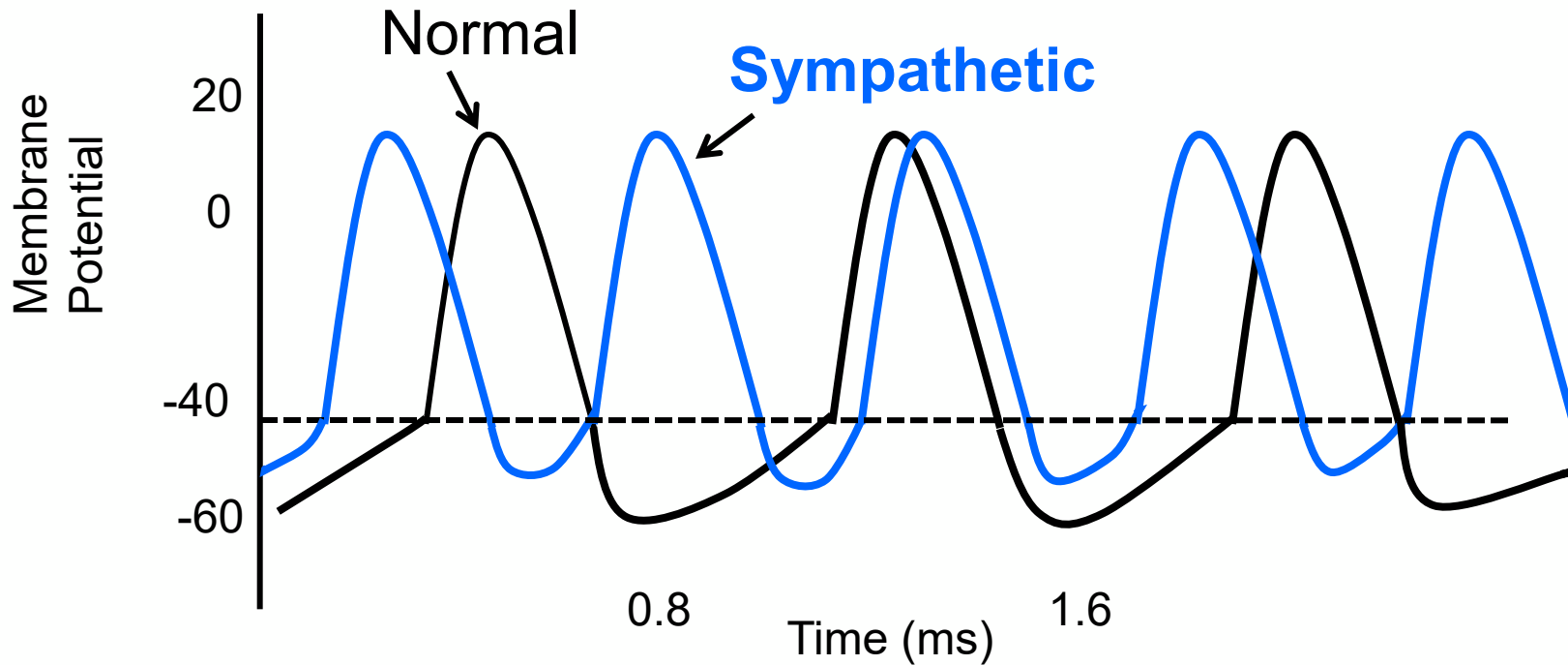
ACh causes a slower rate of rise of the pacemaker potential → ↓ HR
(hyperpolarize and dec ion channel activation)



ACETYLCHOLINE



Effects of Sympathetic Activity on the Heart Rate



Norepinephrine $\uparrow$ rate of depolarization in Phase 4

Effects of Sympathetic Activity on the Heart Rate

NOREPINEPHRINE (β_1 agonist)



Activates cardiac β_1 receptors



$\uparrow$ cAMP



Phosphorylation of i_f (funny) and i_{Ca} channels



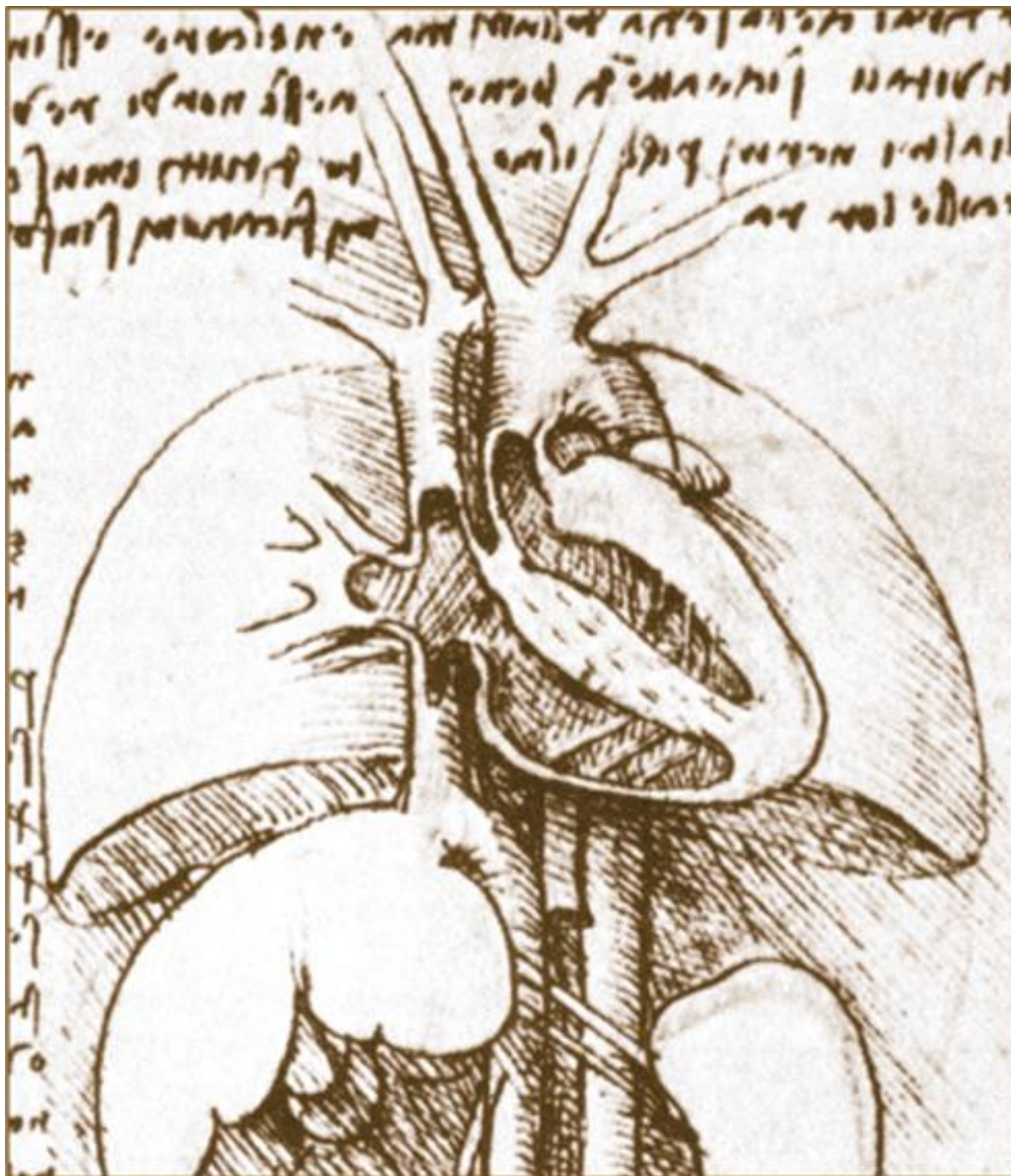
$\uparrow i_f$ and i_{Ca} (inc probability of opening)



$\uparrow$ Rate of depolarization



$\uparrow$ Heart Rate



Source: Valentin Fuster, Robert A. Harrington,
Jagat Narula, Zubin J. Eapen: Hurst's The Heart,
Fourteenth Edition: www.accessmedicine.com
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